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Master's Degree in Computer Science – Artificial Intelligence

Thesis in **Deep Learning**

**Reliable RGB–Thermal Fusion for UAV-Based
Search and Rescue:
Reproducibility, Mechanistic Analysis, and
Architectural Transferability**

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Abstract

Unmanned Aerial Vehicles equipped with visible and thermal cameras can support wilderness Search and Rescue by combining complementary appearance and heat information. Their use, however, raises three coupled problems: either modality may become unavailable or uninformative, the two sensors are not perfectly registered, and the small size of the available data makes architectural comparisons sensitive to training variability and model-selection choices.

This thesis studies RGB–thermal person detection on the WiSARD dataset. It starts from an RT-DETR dual-backbone detector and a Feature Alignment Module (FAM) developed in preceding coursework. The contribution is not the original introduction of FAM, but its reliability-aware evaluation, mechanistic characterization, failure-mode analysis, and transfer to other detector families. An experimental audit revealed that preliminary single-run rankings were not reliable. A locked protocol was therefore defined using five paired training seeds, isolated processes, a fixed ten-epoch horizon, final checkpoints, transferred COCO person heads, and Modal Dropout restricted to training.

On the internal MtErie benchmark, FAM improves additive RT-DETR fusion in all five paired seeds, increasing mean mAP@50 from 0.3080 ± 0.0677 to 0.3780 ± 0.0440 . The main behavioral effect is a higher recall: at confidence 0.25, the mean increase is 0.0990 and is observed in every seed. Direct inspection of offsets, masks, and feature maps confirms that the geometric path is active at P3 and P4, but does not establish physical RGB–IR registration. One checkpoint exhibits a compensated P5 collapse with offsets of approximately 5,030 input pixels and a spatially constant output, despite attaining the highest FAM detection score. A bounded-offset variant removes the observed collapse but reduces mean detection performance, showing that a mechanistically effective correction is not necessarily a better detector.

Further experiments delimit the result. Under the severe scale mismatch of the pre-defined Carnation stress test, FAM is more tolerant than additive fusion, yet visible-only inference remains better than fusion in all five FAM checkpoints. FAM also increases model size and measured detector latency. A five-seed transfer study on a dual-backbone YOLOv10 does not reproduce the RT-DETR gain. The thesis therefore supports FAM as the strongest studied RT-DETR configuration on the internal benchmark, while showing that its benefit, internal behavior, and transferability must be reported with explicit experimental and validity boundaries.

Keywords: RGB–thermal fusion; Search and Rescue; UAV; RT-DETR; YOLOv10; missing modalities; feature alignment; reproducibility.

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Acronyms

AP	Average Precision
CDN	Contrastive DeNoising
COCO	Common Objects in Context
CUDA	Compute Unified Device Architecture
DCNv2	Deformable Convolutional Networks v2
DETR	DEtection TRansformer
FAM	Feature Alignment Module
FN	False Negative
FP	False Positive
IR	Infrared
IoU	Intersection over Union
LFT	Look-Forward-Twice
MQS	Mixed Query Selection
PCA	Principal Component Analysis
RGB	Red, Green, and Blue visible spectrum
SAR	Search and Rescue
SSJ	Stochastic Spatial Jitter
UAV	Unmanned Aerial Vehicle

Chapter 1

Introduction

1.1 Motivation

Wilderness Search and Rescue (SAR) requires locating missing or injured people over large and visually complex areas while time, personnel, visibility, and accessibility are constrained. Unmanned Aerial Vehicles (UAVs) can reduce the area that must be inspected from the ground and provide a viewpoint that is otherwise difficult or dangerous to obtain. Turning aerial imagery into a reliable detection signal is nevertheless challenging: a person may occupy only a small fraction of the frame, vegetation and terrain create clutter, and appearance changes substantially with illumination, altitude, season, and weather.

Visible and thermal infrared cameras provide complementary information. The visible stream contains texture, colour, and scene context, whereas the thermal stream responds to emitted radiation and can retain a useful human signature when visible contrast is weak. WiSARD was introduced specifically to support the study of visual-thermal perception for wilderness SAR across varied acquisition conditions [3]. Complementarity, however, does not imply that using both streams is automatically beneficial. The two cameras are physically distinct, their fields of view and effective scales may differ, and their projections are not perfectly registered. Moreover, either stream may be absent or uninformative in practice. A multimodal detector must therefore learn not only how to combine two signals, but also when and how much to rely on each one.

These deployment concerns interact with an experimental one. The datasets and training budgets available for UAV-based RGB-thermal detection are modest in comparison with general-purpose detection corpora. Small changes in initialization, batch order, stochastic modality masking, or nondeterministic GPU kernels can then lead to visibly different final scores. If architectures are compared through a single execution, the resulting ranking may describe a particular training trajectory rather than a repeatable architectural effect.

1.2 Problem statement

This thesis investigates person detection from paired RGB and thermal imagery on the WiSARD dataset. The main detector family is RT-DETR, adapted to a four-channel

input through two independent backbones and feature-level fusion. The baseline sums corresponding RGB and infrared feature maps at the P3, P4, and P5 pyramid levels. The studied alternative transforms the infrared feature map with a Feature Alignment Module (FAM) before the same additive fusion.

The FAM uses RGB and IR features jointly to predict deformable-convolution offsets and modulation masks. This creates a spatially adaptive transformation of the infrared representation, but the module is not supervised by geometric registration labels. A higher detection score is consequently insufficient to claim that physical sensor registration has been learned. Establishing what the module does requires inspecting its internal variables and intervening on its levels, in addition to measuring the detector output.

The study addresses four connected forms of robustness:

1. **modality robustness**, when RGB or IR is unavailable;
2. **geometric robustness**, when paired streams are imperfectly aligned or differ in scale;
3. **experimental robustness**, when training is repeated under different seeds and nondeterministic execution;
4. **architectural robustness**, when the same fusion idea is moved to another detector family.

1.3 Research questions

The thesis is organized around the following questions.

RQ1 — Experimental reliability. How do run-to-run variability, nondeterministic GPU operations, validation mismatch, checkpoint selection, and process isolation affect the reliability of RGB–IR detector comparisons?

RQ2 — RT-DETR effectiveness and missing modalities. Under a locked paired-seed protocol, does FAM provide a consistent advantage over Additive fusion in RT-DETR, and how does it affect performance when one input modality is unavailable?

RQ3 — Mechanism and failure modes. What geometric-functional transformation does FAM learn across P3, P4, and P5, how stable is it across seeds and sessions, and which pyramid levels contribute to detection performance? Can a smooth bounded-offset constraint remove an observed internal collapse without reducing detection quality?

RQ4 — Operational relevance. How does FAM change the detector’s error profile, its behavior under severe sensor-scale mismatch, and its computational cost?

RQ5 — Architectural transferability. Do the benefits observed with RT-DETR transfer to other detector families under comparable evaluation rules?

1.4 Scope and relationship with preceding course-work

This thesis continues two preceding course projects. The Deep Learning project studied DETR, Modal Dropout, RT-DETR Additive fusion, fixed tiling, and CMX. The Computer Vision project subsequently introduced the FAM on RT-DETR, designed a Hybrid CMX variant, and adapted Deformable DETR to channel fusion with FAM. These components define the starting point of the present work.

The FAM is therefore *not* presented as a new architecture invented in this thesis. It is described in full so that the document remains self-contained, but its provenance is stated explicitly. The present contribution is the reliability-aware validation of that module: its performance across paired runs, its internal behavior, its failure modes, the effect of a targeted correction, and the extent to which the result transfers outside the original RT-DETR setting.

The distinction also applies to the evidence. Early experiments were useful for discovering bugs and generating hypotheses, but many were individual runs produced before the final validation and checkpoint policy was defined. They are retained as an exploratory development history and are not mixed with the main multi-seed estimates.

1.5 Contributions

The contributions developed after the preceding course projects are:

1. an audit of variability, validation composition, checkpoint policy, model initialization, GPU determinism, and process isolation;
2. a locked RT-DETR comparison of six configurations over five paired seeds, including initialization and direct-warp ablations;
3. a multi-seed diagnostic study of offsets, masks, feature statistics, PCA projections, and session dependence;
4. a factorial post-hoc ablation of FAM activity at P3, P4, and P5;
5. the identification of a compensated P5 collapse and a five-seed test of a smooth bounded-offset correction;
6. a paired error analysis across confidence thresholds, including size-stratified recall and predefined qualitative examples;
7. a frozen one-shot stress test on the Carnation acquisition and a controlled detector-only compute benchmark;
8. a complete DINO implementation with Contrastive DeNoising, Mixed Query Selection, and Look-Forward-Twice, reported as a negative exploratory result for the studied configuration;
9. an extension of the Deformable DETR line with FAM and Stochastic Spatial Jitter, retained as exploratory evidence;

10. a dual-backbone YOLOv10 implementation with Additive or FAM fusion and feature-level modality gating, evaluated under a five-seed final protocol.

Negative results are part of these contributions. They delimit which mechanistic changes improve internal behavior without improving detection and which RT-DETR findings fail to transfer under the studied conditions.

1.6 Thesis organization

Chapter 2 introduces the dataset, detector families, multimodal fusion, missing-modality training, and the inherited architectural starting point. Chapter 3 reconstructs the post-coursework exploratory development and explains how the variability investigation led to the final protocol. Chapter 4 presents the main RT-DETR results, internal FAM diagnostics, bounded-offset intervention, error analysis, Carnation stress test, computational cost, and YOLOv10 transfer study. Finally, Chapter 5 answers the research questions, states the validity boundaries, and outlines future work.

Chapter 2

Background and Research Starting Point

2.1 UAV-based RGB–thermal detection for Search and Rescue

Person detection in aerial wilderness imagery differs from conventional street-level pedestrian detection. Subjects are frequently small, partially occluded, and observed from changing altitudes and viewing angles. Natural backgrounds provide repeated textures, warm rocks may resemble thermal human signatures, and targets may exhibit weak visible or thermal contrast. The WiSARD dataset addresses this setting with labeled visual and thermal imagery collected from UAV flights over diverse terrains and acquisition conditions [3].

RGB–thermal sensing seeks to reduce dependence on a single observation channel. Multispectral pedestrian benchmarks have long shown the potential of combining visible and thermal cues [5], while the broader multimodal-learning literature emphasizes both complementarity and the difficulty of learning balanced representations [2, 9]. In this thesis, robustness is evaluated explicitly in three input conditions: full fusion (VIS+IR), visible-only (VIS), and infrared-only (IR).

2.2 Object detection and evaluation

An object detector maps an image to a set of class scores and bounding boxes. For a predicted box B_p and a ground-truth box B_g , Intersection over Union is

$$\text{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}. \quad (2.1)$$

Average Precision (AP) integrates precision as a function of recall after ranking detections by confidence. The primary metric in the final experiments is mAP@50, for which a match requires IoU at least 0.50. COCO mAP averages AP over IoU thresholds from 0.50 to 0.95 and provides a stricter localization summary [6]. mAP@75 and mean Average Recall (mAR) are retained as secondary metrics.

The confidence value used to collect detections is not itself the IoU match criterion.

A low confidence threshold preserves candidates for AP computation, but may be unusable as an operational alarm threshold. This distinction is important in Section 4.3, where threshold 0.01 produces high recall and tens of false positives per image.

2.3 Detection Transformer families

2.3.1 DETR

DETR formulates detection as direct set prediction [4]. A convolutional backbone produces image features, a Transformer encoder–decoder processes them together with a fixed number of object queries, and a bipartite matching loss assigns predictions to ground-truth objects. This removes the need for anchor design and non-maximum suppression in the original model, but training converges slowly and small-object performance is challenging.

The Deep Learning project that precedes this thesis used DETR to establish the value of stochastic missing-modality training. DETR is not rerun in the final campaign: its role is to motivate the training regime inherited by later architectures.

2.3.2 Deformable DETR

Deformable DETR replaces dense attention over all spatial locations with multi-scale deformable attention around learned reference points [14]. Each query attends to a small set of sampled locations across multiple feature levels, improving convergence and making multi-scale processing tractable. In this project, however, its CUDA backward path also exposed substantial run-to-run variability. The Deformable DETR experiments are therefore treated as exploratory because they predate the locked RT-DETR checkpoint and initialization protocol.

2.3.3 DINO

DINO extends DETR-like detectors through three principal mechanisms: Contrastive DeNoising (CDN), Mixed Query Selection (MQS), and Look-Forward-Twice (LFT) box refinement [10]. The project initially contained a legacy CDN variant that was historically labeled DINO, although MQS and effective LFT were absent. A complete implementation was developed after the course projects and is documented separately in Section 3.3.2. Its low base performance under the studied budget prevented an informative follow-up with FAM; this is a configuration-specific negative result, not a general assessment of DINO.

2.3.4 RT-DETR

RT-DETR targets end-to-end real-time detection with an efficient hybrid encoder and uncertainty-minimal query selection [12]. Its encoder separates within-scale interaction from cross-scale fusion, reducing the cost of processing multi-scale features. The implementation used here starts from the pretrained RT-DETR R50vd checkpoint and exposes three backbone outputs, denoted P3, P4, and P5 with strides 8, 16, and 32.

The term “real-time” is not inferred from the architecture name. This thesis reports latency for the detector forward pass on one specified GPU and excludes preprocessing and postprocessing. Any broader real-time claim would require an end-to-end deployment measurement.

2.3.5 YOLOv10

YOLOv10 is an end-to-end member of the YOLO family that combines consistent dual assignments with an efficiency-oriented architecture [8]. It provides a structurally different transfer target from RT-DETR: the feature extractor and FPN/PAN neck are convolutional, while detection uses the YOLOv10 one-to-many and one-to-one training branches. The thesis adapts YOLOv10-s to two modality-specific backbones and fuses their P3, P4, and P5 outputs before the shared neck.

2.4 Multimodal fusion

Fusion methods can be grouped by where the modalities interact. Early fusion combines inputs or low-level representations, late fusion combines independent predictions, and intermediate fusion combines learned features. This work uses intermediate fusion because each sensor receives a dedicated backbone while the detector still operates on a joint multi-scale representation.

The inherited RT-DETR baseline uses element-wise addition:

$$F_{\text{fused}}^{(i)} = F_{\text{RGB}}^{(i)} + F_{\text{IR}}^{(i)}, \quad i \in \{P3, P4, P5\}. \quad (2.2)$$

Addition introduces no trainable fusion parameters and preserves feature-map shape, but assumes that corresponding cells are meaningfully compatible. A sensor displacement or scale difference violates this assumption. More complex interaction does not automatically solve the problem: multimodal networks can overfit or optimize their modalities at different rates [9].

The preceding projects also examined CMX, a cross-modal rectification and fusion framework originally proposed for RGB-X semantic segmentation [11]. The specific RT-DETR integration was not competitive on WiSARD. This finding is retained as an exploratory result of that implementation, not as a general rejection of attention-based fusion.

2.5 Missing modalities and Modal Dropout

ModDrop was proposed to preserve modality-specific representations by randomly dropping input channels during multimodal training [7]. The project applies the same principle at the modality level. For a paired sample $(x_{\text{RGB}}, x_{\text{IR}})$, training draws one of three conditions:

$$\tilde{x} = \begin{cases} (x_{\text{RGB}}, 0), & \text{with probability } p_{\text{VIS}}, \\ (0, x_{\text{IR}}), & \text{with probability } p_{\text{IR}}, \\ (x_{\text{RGB}}, x_{\text{IR}}), & \text{with probability } p_{\text{fusion}}. \end{cases} \quad (2.3)$$

The inherited schedule uses $p_{\text{VIS}} = 0.2$, $p_{\text{IR}} = 0.2$, and $p_{\text{fusion}} = 0.6$. It is a motivated exploratory choice from the preceding DETR study, not a statistically demonstrated optimum. A software audit conducted for this thesis found that modality dropout was also being applied during validation and test. The pipeline was corrected so that the sampling in Equation (2.3) is active only during training.

2.6 Feature-level geometric transformation

2.6.1 Deformable convolution

Standard convolution samples a fixed grid $\mathcal{R}$ around each spatial position p . Modulated deformable convolution augments every kernel point p_k with an input-dependent offset Δp_k and a modulation value m_k [13]:

$$y(p) = \sum_{k \in \mathcal{R}} w_k x(p + p_k + \Delta p_k) \sigma(m_k). \quad (2.4)$$

Sampling at non-integer positions uses bilinear interpolation. The operation therefore combines a geometric sampling change with learned filtering and channel mixing. Its output cannot be interpreted as a pure warp unless those additional effects are controlled.

2.6.2 Inherited Feature Alignment Module

For every pyramid level, the FAM concatenates RGB and IR features and predicts 18 offset channels and 9 mask logits for a 3×3 DCNv2 kernel:

$$[\Delta, m] = \text{Conv}_{3 \times 3}([F_{\text{RGB}}, F_{\text{IR}}]), \quad (2.5)$$

$$\tilde{F}_{\text{IR}} = \text{DCNv2}(F_{\text{IR}}, \Delta, \sigma(m)), \quad (2.6)$$

$$F_{\text{fused}} = F_{\text{RGB}} + \tilde{F}_{\text{IR}}. \quad (2.7)$$

The offset/mask predictor is initialized to zero, so initial offsets are null and masks are approximately 0.5. The DCNv2 filter weights retain their own initialization. Consequently, the standard FAM does *not* start as a mathematical identity: null offsets preserve the initial sampling geometry but do not make the filtered output equal to the input.

Because there is no ground-truth displacement field, the thesis uses the term *geometric-functional transformation*. Nonzero offsets show that the geometric path is active; they do not prove recovery of the physical mapping between the cameras. This distinction motivates the internal diagnostics in Section 4.2.

2.7 Dataset organization and evaluation conditions

All final comparisons use synchronized WiSARD RGB–IR pairs. The effective split audit is summarized in Table 2.1.

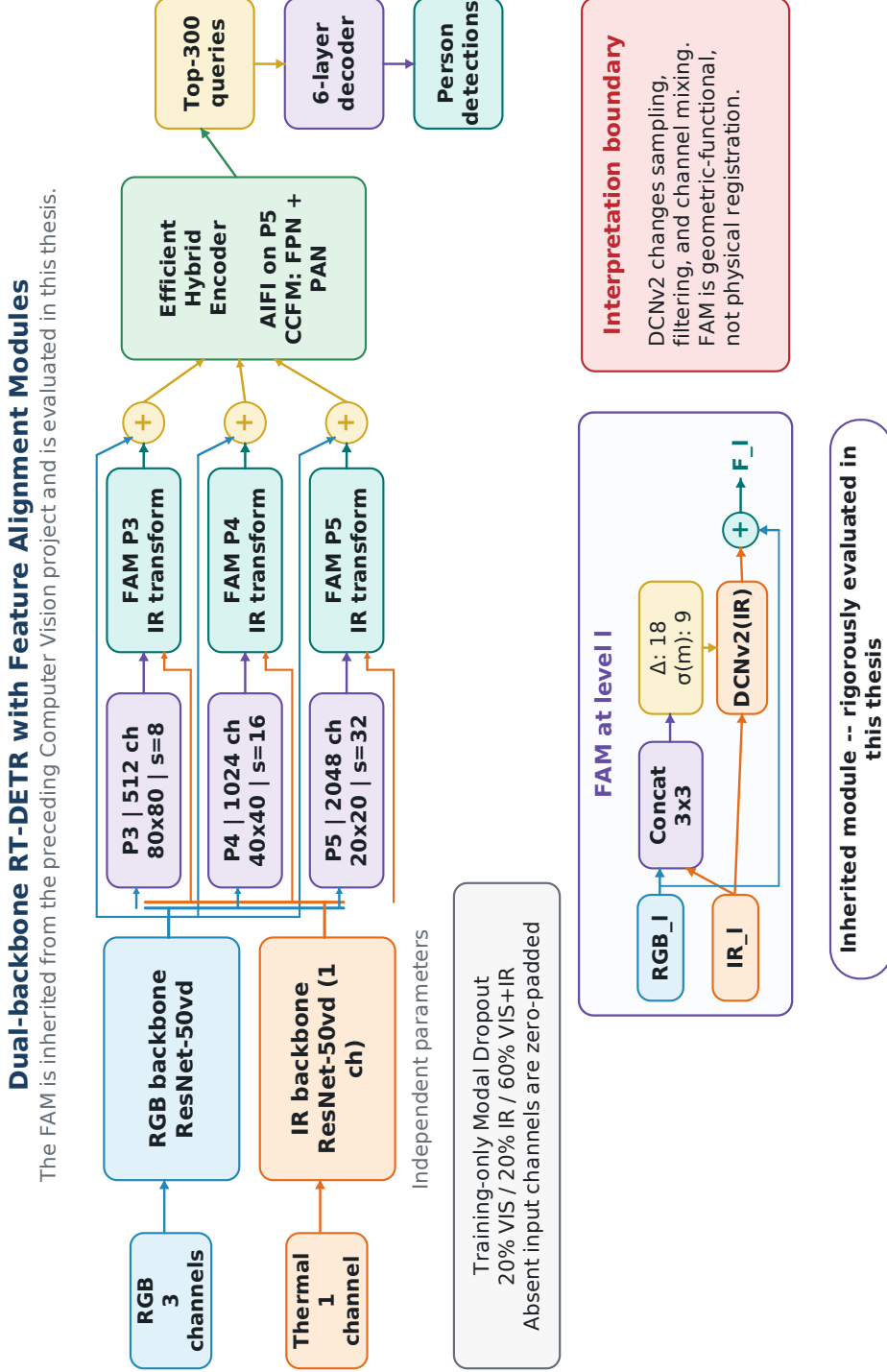


Figure 2.1: Dual-backbone RT-DETR and the inherited Feature Alignment Module. At each pyramid level, RGB and IR features jointly predict 18 offset components and nine modulation masks; DCNv2 then transforms the IR feature before additive fusion. In the final data path, training-only Modal Dropout zero-pads the missing input channels while preserving the four-channel interface. The FAM is the starting point inherited from the preceding Computer Vision project, not a new architecture introduced by this thesis.

Table 2.1: Effective dataset composition after the project filters. Median box area is expressed as a percentage of the evaluation image area.

Split	Sessions	Frames	Empty frames	Boxes/frame	Median box area
Train	3	4,022	17.85%	2.464	0.1641%
Validation	1	273	67.40%	0.542	0.0133%
Internal test	3	708	2.68%	2.500	0.0764%

The validation set contains one FHL session, 184 empty frames, and 148 boxes. Its median target area is approximately twelve times smaller than in training. It is informative as evidence of a strong distribution shift, but it is not a representative selector for performance on the three MtErie sequences.

MtErie is called the *internal development benchmark*. It was consulted during architectural development and is not an untouched holdout. Final paired comparisons remain useful, particularly after the protocol was frozen, but may be optimistic with respect to a new acquisition. Carnation is evaluated once as a predefined external stress case and is not substituted for a general domain-generalization benchmark.

At evaluation time, the three modality conditions retain a common dataset and annotation set:

VIS+IR

both synchronized streams are provided;

VIS the infrared input or branch contribution is disabled according to the architecture’s modality contract;

IR the visible input or branch contribution is disabled.

The exact missing-modality path differs between RT-DETR and YOLOv10 and is reported with their results rather than assumed to be architecturally equivalent.

2.8 Inherited experimental starting point

The Deep Learning project established DETR with Modal Dropout, the RT-DETR Additive baseline, a fixed 2×2 tiling experiment related to slicing-aided small-object detection [1], and the initial CMX comparison. The Computer Vision project introduced FAM on RT-DETR, Hybrid CMX, and Deformable DETR with nonlinear channel fusion and FAM. Their historical scores were based primarily on the earlier checkpoint policy and, for several RT-DETR variants, on individual executions.

Those results generated the central hypothesis that transforming IR features before fusion could be preferable to direct addition. They did not yet answer whether the advantage was consistent across runs, whether the learned offsets were geometrically meaningful, whether all feature levels behaved correctly, or whether the result transferred to another detector. These questions define the point at which the present thesis begins.

Chapter 3

Exploratory Development and Experimental Methodology

3.1 Two phases of evidence

The project contains two qualitatively different forms of evidence. During the *exploratory phase*, individual runs were used to expose implementation problems, compare broad architectural directions, and generate new hypotheses. This phase produced the evolution of the FAM, the SSJ regularizer, extensions to Deformable DETR, and a complete DINO implementation. Its numerical results describe the development history but do not estimate run-to-run variability under a common final protocol.

The subsequent *locked-protocol phase* was motivated by the observation that repeated executions could change the ranking of configurations. In this phase, the main RT-DETR hypotheses were evaluated with five paired seeds, fixed training duration, final checkpoints, isolated processes, consistent initialization, and aggregate reporting. The distinction is retained throughout the thesis: exploratory evidence motivates a test, whereas locked-protocol evidence supports the principal comparison.

3.2 Post-coursework evolution of the FAM

3.2.1 Lazy registration bug

The first RT-DETR FAM implementation instantiated its alignment layers lazily during the first forward pass. The optimizer was created earlier and therefore did not contain those parameters. The offset predictor remained at zero, while the deformable-convolution weights remained fixed at their random initialization. This model did not implement an identity transformation and did not train a functional FAM, even though its fusion score was high in one run.

The extremely low IR-only score of that checkpoint is consistent with a weak or unusable thermal path, but the final metrics do not prove literal cancellation of the IR backbone. The run is useful as a debugging case, not as a competitive model.

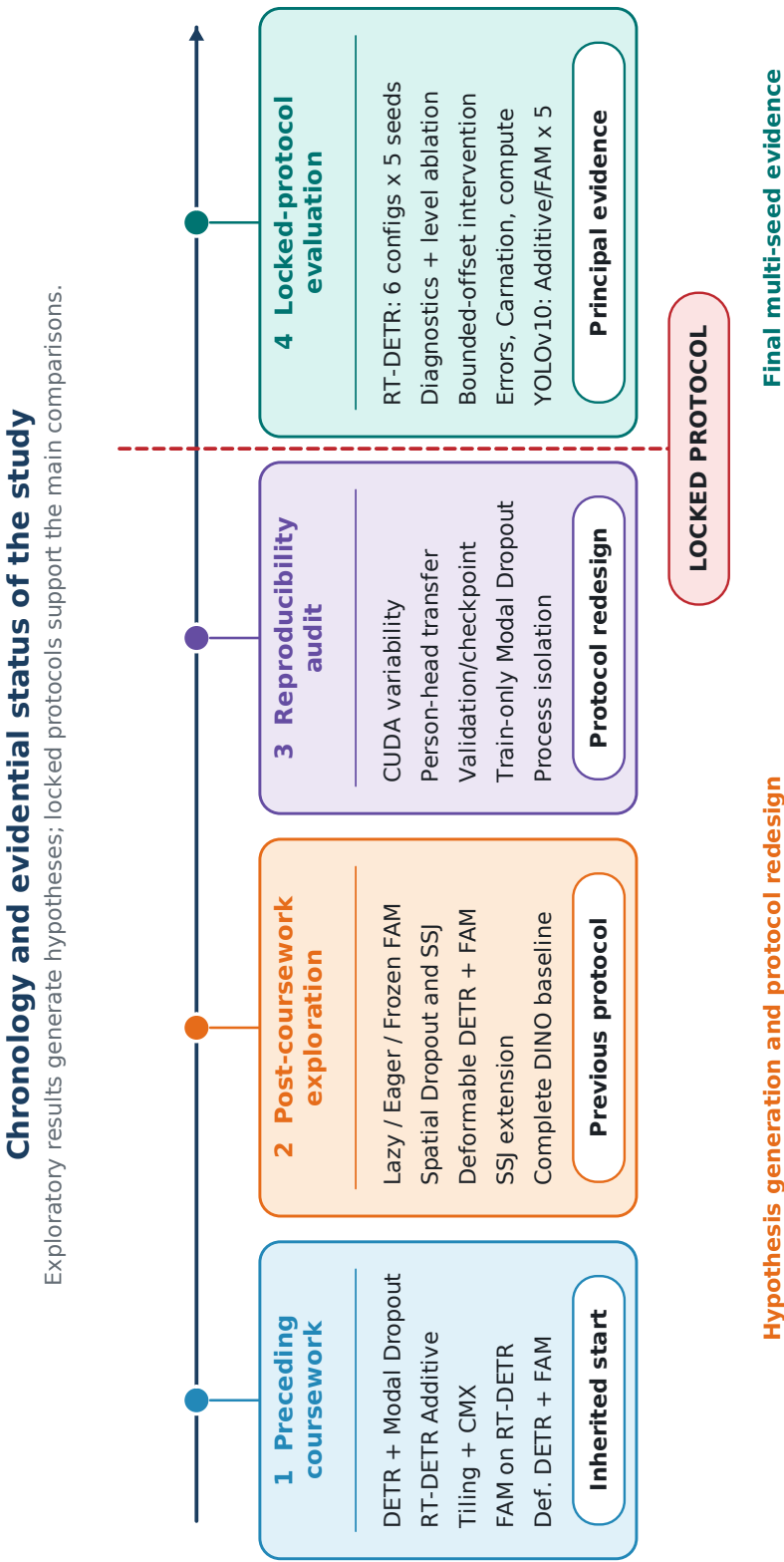


Figure 3.1: Chronology and evidential status of the study. The first two stages establish the inherited starting point and record hypothesis-generating work conducted under the previous protocol. The audit motivates the redesign; the vertical boundary marks the transition to the locked RT-DETR and YOLOv10 multi-seed evaluations. Card widths encode stages, not elapsed time.

3.2.2 Eager, Frozen, and Spatial Dropout variants

The implementation was corrected by registering all FAM layers in the model constructor before optimizer creation. This *Eager FAM* allowed the alignment parameters to train from the first optimization step. A Frozen variant then tested a fixed, eagerly registered transform: gradients could propagate through it to the IR backbone, but the FAM parameters themselves did not change. A further variant applied channel-wise spatial dropout to the transformed IR features immediately before addition.

These interventions did not isolate a single causal explanation for the historical Lazy score. In particular, the Lazy bug generated a fixed random filter with null offsets, not stochastic geometric jitter. Their role was to motivate a cleaner hypothesis about perturbing sampling geometry during training.

3.2.3 Stochastic Spatial Jitter

Stochastic Spatial Jitter adds zero-mean Gaussian noise to the predicted offsets only during training:

$$\Delta_{\text{train}} = \Delta_{\text{FAM}} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2). \quad (3.1)$$

Unlike feature dropout, SSJ does not explicitly remove channels. It perturbs the positions sampled by DCNv2 around the correction currently predicted by the FAM. At inference, ε is absent. The intended regularization is therefore local uncertainty around a learned transformation, rather than a return to the original sensor displacement.

Table 3.1 reports the single-run results that generated the SSJ hypothesis. They are deliberately separated from the final comparison.

Table 3.1: Historical single-run FAM development results. These values belong to the exploratory phase and do not determine the final model selection.

Variant	VIS	IR	VIS+IR
Lazy registration bug	0.3000	0.0054	0.4286
Eager FAM	0.2620	0.2629	0.3960
Frozen FAM	0.2517	0.2250	0.3857
Eager FAM + 40% Spatial Dropout	0.2716	0.2248	0.3936
Eager FAM + SSJ	0.2697	0.1818	0.4381

The SSJ run reached the largest historical fusion value, but its apparent advantage did not survive paired replication: the locked-protocol result in Section 4.1 selects the standard FAM without SSJ.

3.3 Exploratory architectural extensions

3.3.1 Deformable DETR with channel fusion, FAM, and SSJ

The inherited Deformable DETR adaptation uses independent ResNet-50 RGB and IR backbones. The IR input stem is initialized by averaging the three RGB input channels. At the three native backbone levels, corresponding features are concatenated along the channel dimension and passed through a nonlinear fusion block:

$$F_{\text{fused}}^{(i)} = \text{ReLU}\left(\text{GN}_{32}\left(\text{Conv}_{1 \times 1}([F_{\text{RGB}}^{(i)}, F_{\text{IR}}^{(i)}])\right)\right). \quad (3.2)$$

This preserves the spatial dimensions required by the pretrained attention reference points. It replaced an unsuccessful spatial-concatenation attempt that doubled feature-map height, forced a reduction in pyramid levels because of memory use, and produced near-zero detection performance.

The Computer Vision project had already integrated FAM before the channel fusion block. The post-coursework extension added SSJ under the same previous experimental regime. Five runs are available for each configuration, but they precede the final RT-DETR head initialization, checkpoint rule, and validation policy. Their median and range are reported in Table 3.2 as exploratory transfer evidence.

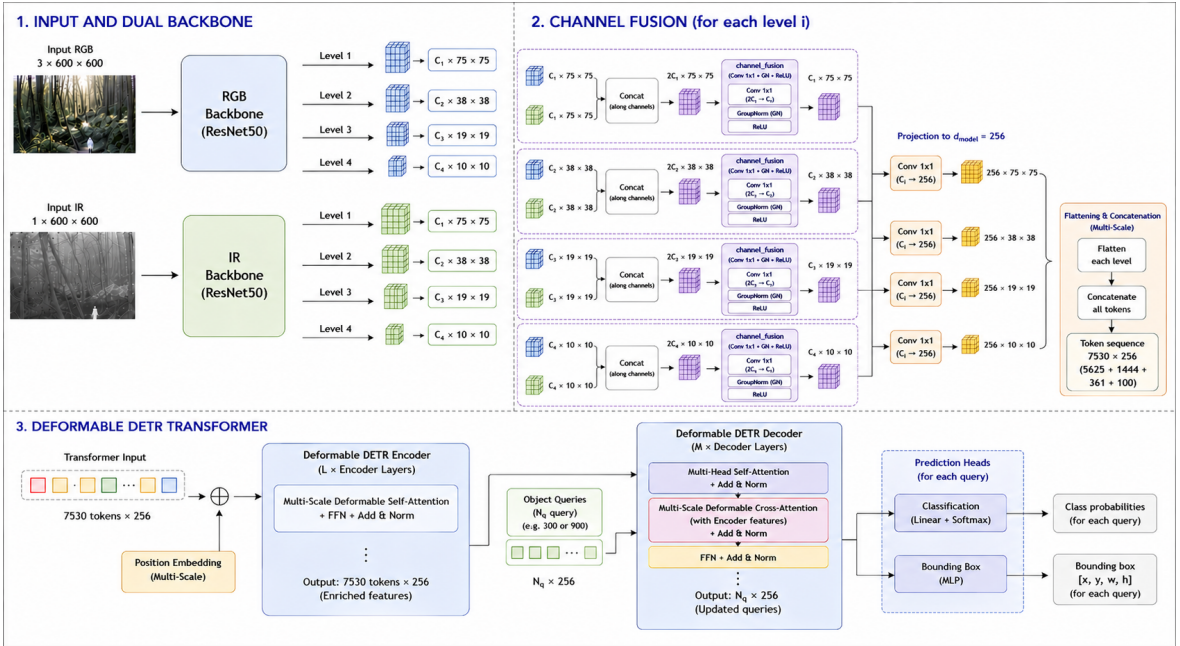


Figure 3.2: Deformable DETR dual-backbone fusion architecture. The fourth encoder level is derived from the fused deepest native feature and does not receive a separate fourth FAM. This diagram documents an exploratory line that precedes the final RT-DETR protocol.

FAM has the highest observed median, but the ranges overlap substantially. SSJ does not reproduce the gain suggested by its historical RT-DETR run and has the widest VIS+IR range. A definitive cross-architecture ranking would require retraining these models under a protocol aligned with the final RT-DETR study.

Table 3.2: Exploratory Deformable DETR results, reported as median [minimum–maximum] mAP@50 over five runs.

Configuration	VIS	IR	VIS+IR
Deformable DETR	0.157 [0.131–0.197]	0.115 [0.086–0.142]	0.249 [0.199–0.303]
+ FAM	0.184 [0.171–0.253]	0.131 [0.097–0.165]	0.294 [0.213–0.314]
+ FAM + SSJ	0.142 [0.115–0.206]	0.088 [0.073–0.109]	0.263 [0.170–0.350]

3.3.2 Complete DINO implementation

The complete DINO implementation retains the dual-backbone and channel-fusion base of Deformable DETR and activates all three defining mechanisms. MQS uses the top encoder proposals as spatial anchors while drawing query content from an independent learned embedding. LFT keeps the refined reference passed to the next attention layer detached while preserving a graph-connected copy for the following prediction loss. CDN inserts noisy positive and negative target queries during training and applies the required decoder attention mask.

Earlier runs labeled “DINO” in project reports were a CDN legacy variant: box refinement was disabled, MQS was absent, and the CDN mask did not reach the decoder self-attention. They are not ablations of the complete model and are excluded from the main conclusion.

The complete base was trained five times and produced the values in Table 3.3. Since even its maximum fusion result remained below the median of the historical Deformable DETR base, complete FAM and SSJ variants were not launched without a new diagnostic hypothesis.

Table 3.3: Complete DINO base result under the previous protocol, reported as median [minimum–maximum] mAP@50 over five runs.

Configuration	VIS	IR	VIS+IR
Complete DINO base	0.0565 [0.0232–0.1091]	0.0881 [0.0671–0.1473]	0.1088 [0.0669–0.1635]

This is a negative result for the implemented configuration and training budget. It does not show that DINO is intrinsically unsuitable for WiSARD.

3.4 Reproducibility investigation

3.4.1 Why single runs were insufficient

Two native-CUDA RT-DETR training runs with the same nominal seed and pretrained person head produced final test mAP@50 values of 0.4171 and 0.2934. Their first forward loss was identical, but their weight hashes diverged after the first backward pass. A difference of 0.1236 is large enough to reverse many of the historical architectural comparisons.

The investigation therefore recorded runtime fingerprints, initialization hashes, sample indices, modality-dropout decisions, input hashes, losses, and post-update weight hashes. Native execution diverged at the first optimizer step even when inputs and the initial loss matched. A deterministic interpolation substitute for the deformable-attention sampling path produced identical traces, metrics, and final checkpoint hashes in two complete RT-DETR runs. This establishes that strict reproduction is possible for that substitute path, while native RT-DETR CUDA is not fully deterministic.

3.4.2 Pretrained single-class head

The original COCO checkpoint has 80 classes, whereas the task has one class, *person*. Randomly initialized classification and denoising heads made the seed affect a substantial part of the initial model. The final setup transfers the COCO person row into all compatible RT-DETR classification heads and the denoising embedding. This makes the initial model hash independent of the experiment seed and reduces, but does not eliminate, amplification of native CUDA perturbations.

In 20-step probes, the mean range of the loss across identical-seed native runs decreased from 4.0763 with random heads to 0.4393 with the transferred person head. The maximum range decreased from 20.8075 to 1.6424. The change is therefore a mitigation and a more controlled transfer-learning initialization, not a determinism guarantee.

3.4.3 DCNv2 limitation

The strict FAM probe stops at the first backward pass because the CUDA backward of `torchvision.ops.DeformConv2d` has no deterministic implementation in the studied environment. A numerically equivalent PyTorch fallback was too slow at the 512, 1024, and 2048 channel feature levels for full training. FAM must consequently be evaluated statistically under native CUDA rather than through a strictly deterministic full campaign.

3.4.4 Modal Dropout and evaluation leakage

The data pipeline originally sampled Modal Dropout during validation and test as well as training. This made evaluation inputs stochastic and inconsistent with the intended VIS, IR, and VIS+IR conditions. The loader was corrected and regression-tested so that dropout is active only in training. All final evaluations disable it.

3.4.5 Validation and checkpoint policy

The effective validation split contains 273 FHL frames, of which 184 are empty, and only 148 boxes. Its objects are substantially smaller than those in training and MtErie. Near-zero validation metrics are therefore evidence of a domain and scale mismatch, not proof that the validator is defective or that the split has no information.

Using that split for early stopping does not provide a representative rule for MtErie. Conversely, choosing an epoch from MtErie would tune directly on the internal test. The final RT-DETR protocol resolves the immediate comparison by fixing ten epochs in

advance and evaluating the final `latest` checkpoint. Validation is not run as a selector. This does not replace the need for a future representative train-derived validation set.

3.4.6 Process isolation

Sequential grids in one Python process showed progressive slowdowns and could retain allocator or framework state. Each seed in the final campaigns is therefore launched in a new process. The same principle is applied to compute benchmark trials, where an initial in-process memory result was rejected after cumulative allocations were detected.

3.5 Locked RT-DETR protocol

The final RT-DETR comparison uses six independently trained configurations:

1. Additive fusion;
2. standard FAM with DCNv2;
3. FAM with IR feature dropout probability 0.4;
4. FAM with SSJ standard deviation 0.5;
5. an identity-initialized DCNv2 FAM;
6. a direct bilinear Grid Sample warp.

The identity DCNv2 ablation retains the standard FAM structure but initializes the central kernel as a channel-wise identity with weight two, exactly compensating the initial mask value 0.5. The Grid Sample variant predicts one two-dimensional displacement per spatial cell and applies a bilinear warp without a learned post-sampling convolution. It is exactly the identity at zero offset through a correction that subtracts numerical sampling of the base grid.

The same seed index is paired across configurations to reduce the noise in architectural contrasts. It does not make native executions deterministic.

3.6 Statistical reporting

For every final configuration, the thesis reports all five seed values, mean, median, sample standard deviation, minimum–maximum range, and a Student t -interval for the mean. Architectural contrasts use within-seed differences, their mean and interval, and the number of paired wins. Paired t -tests and exact Wilcoxon signed-rank tests are included as exploratory summaries.

With only five pairs, intervals are imprecise, p-values are discrete or sensitive to assumptions, and no correction is applied for the many possible comparisons. Conclusions therefore prioritize effect size, sign consistency, and agreement across metrics over a binary significance threshold.

Table 3.4: Locked RT-DETR protocol used by all six final configurations.

Component	Fixed choice
Training seeds	Paired seeds 40–44; five runs per configuration
Execution	Native CUDA; one isolated Python process per seed
Training horizon	10 fixed epochs
Checkpoint	Final latest ; no test-based or validation-based selection
Initialization	Pretrained RT-DETR R50vd and transferred COCO person heads
Modal Dropout	VIS/IR/fusion probabilities 0.2/0.2/0.6, training only
Validation	Not used during training or for checkpoint selection
Primary benchmark	MtErie internal development benchmark
Primary metric	mAP@50
Evaluation	VIS, IR, and VIS+IR with a fixed evaluation seed

The experimental unit for architectural comparison is the independently trained run/checkpoint. Frames, bounding boxes, samples used in a diagnostic, feature-map cells, and the nine DCNv2 kernel points are nested observations, not independent replicates. Diagnostics first aggregate within a checkpoint and only then compare the five checkpoints.

Chapter 4

Experimental Evaluation

4.1 Main RT-DETR multi-seed study

4.1.1 Raw and aggregate VIS+IR results

All thirty training runs completed the fixed ten-epoch protocol and produced a final `latest` checkpoint. Table 4.1 reports the primary VIS+IR mAP@50 result for every experimental unit.

Table 4.1: Final RT-DETR VIS+IR mAP@50 by paired training seed.

Seed	Additive	FAM	FAM + IR Dropout	FAM + SSJ	Identity DCNv2	Grid Sample
40	0.2566	0.3783	0.4087	0.3663	0.4336	0.4050
41	0.4029	0.4335	0.3234	0.3734	0.2113	0.3483
42	0.2958	0.3129	0.4452	0.3528	0.2926	0.4083
43	0.3476	0.3964	0.3986	0.4030	0.2860	0.3326
44	0.2372	0.3690	0.3598	0.3792	0.4162	0.3564

The aggregate distributions in Table 4.2 show why a single run is insufficient. Identity DCNv2 ranges from 0.2113 to 0.4336, and the highest value of several individual seeds belongs to a configuration that is not best on average or consistently paired with the reference.

Table 4.2: Aggregate final RT-DETR VIS+IR mAP@50. Standard deviations are sample standard deviations across five checkpoints.

Configuration	Mean	Median	SD	Min-max	95% CI of mean
Additive	0.3080	0.2958	0.0677	0.2372–0.4029	0.2239–0.3921
FAM	0.3780	0.3783	0.0440	0.3129–0.4335	0.3234–0.4326
FAM + IR Dropout	0.3871	0.3986	0.0469	0.3234–0.4452	0.3289–0.4453
FAM + SSJ	0.3749	0.3734	0.0185	0.3528–0.4030	0.3519–0.3980
Identity DCNv2	0.3280	0.2926	0.0943	0.2113–0.4336	0.2109–0.4451
Grid Sample	0.3701	0.3564	0.0345	0.3326–0.4083	0.3273–0.4129

FAM increases mean mAP@50 over Additive by 0.0700 and wins in all five paired seeds. Its mean COCO mAP also increases from 0.1059 ± 0.0243 to 0.1351 ± 0.0096 , mAP@75 from 0.0474 ± 0.0128 to 0.0692 ± 0.0043 , and mAR@100 from 0.2474 ± 0.0302 to 0.2798 ± 0.0141 . The direction is therefore not limited to the primary IoU threshold.

4.1.2 Paired contrasts and model selection

Table 4.3 compares FAM with Additive and compares each FAM regularization or ablation with the standard FAM. With five pairs, the tests and intervals are exploratory and are not used as a binary decision rule.

Table 4.3: Paired VIS+IR mAP@50 contrasts. A win means that the first configuration exceeds its reference for the same training seed.

Contrast	Mean delta	SD delta	95% CI	Wins	Paired (t) / Wilcoxon (p)
FAM – Additive	+0.0700	0.0531	[+0.0040,+0.1359]	5/5	0.0421 / 0.0625
FAM + IR Dropout – FAM	+0.0091	0.0869	[-0.0988,+0.1171]	3/5	0.8260 / 0.8125
FAM + SSJ – FAM	-0.0031	0.0369	[-0.0488,+0.0427]	3/5	0.8617 / 1.0000
Identity DCNv2 – FAM	-0.0501	0.1170	[-0.1953,+0.0952]	2/5	0.3928 / 0.6250
Grid Sample – FAM	-0.0079	0.0724	[-0.0979,+0.0820]	2/5	0.8192 / 1.0000

IR Dropout has the highest raw mean, but its mean improvement over FAM is only 0.0091, changes sign across seeds, and has a wide interval. It is not selected as a reliable improvement. SSJ has the smallest descriptive standard deviation in the six-configuration campaign but does not increase mean performance. Five runs are insufficient to infer that it reduces population variance.

Identity initialization does not solve final instability. Grid Sample is close to standard FAM, but its parameterization is not equivalent to DCNv2: it predicts one displacement per cell and performs no post-sampling filtering or channel mixing. Its result supports the relevance of a geometric component, but does not causally assign all FAM improvement to physical alignment.

The main model for subsequent analysis is therefore standard FAM with DCNv2, without SSJ and without IR Dropout. The decision favors the consistent paired gain over Additive rather than the highest numerical mean.

4.1.3 Missing-modality evaluation

Each final checkpoint was evaluated in VIS, IR, and VIS+IR with a fixed evaluation seed and Modal Dropout disabled. The aggregate mAP@50 results are reported in Table 4.4.

VIS+IR exceeds the better single modality of the same checkpoint in all thirty cases. Depending on the configuration, the mean within-checkpoint margin is between 0.1067 and 0.1454. This is the most uniform result of the main campaign: on MtErie, complementary information remains useful even though the individual modality scores vary substantially.

FAM also improves Additive in VIS-only in 5/5 seeds, with a mean delta of 0.0899, and in IR-only in 4/5, with a mean delta of 0.0462. In RT-DETR, a missing input

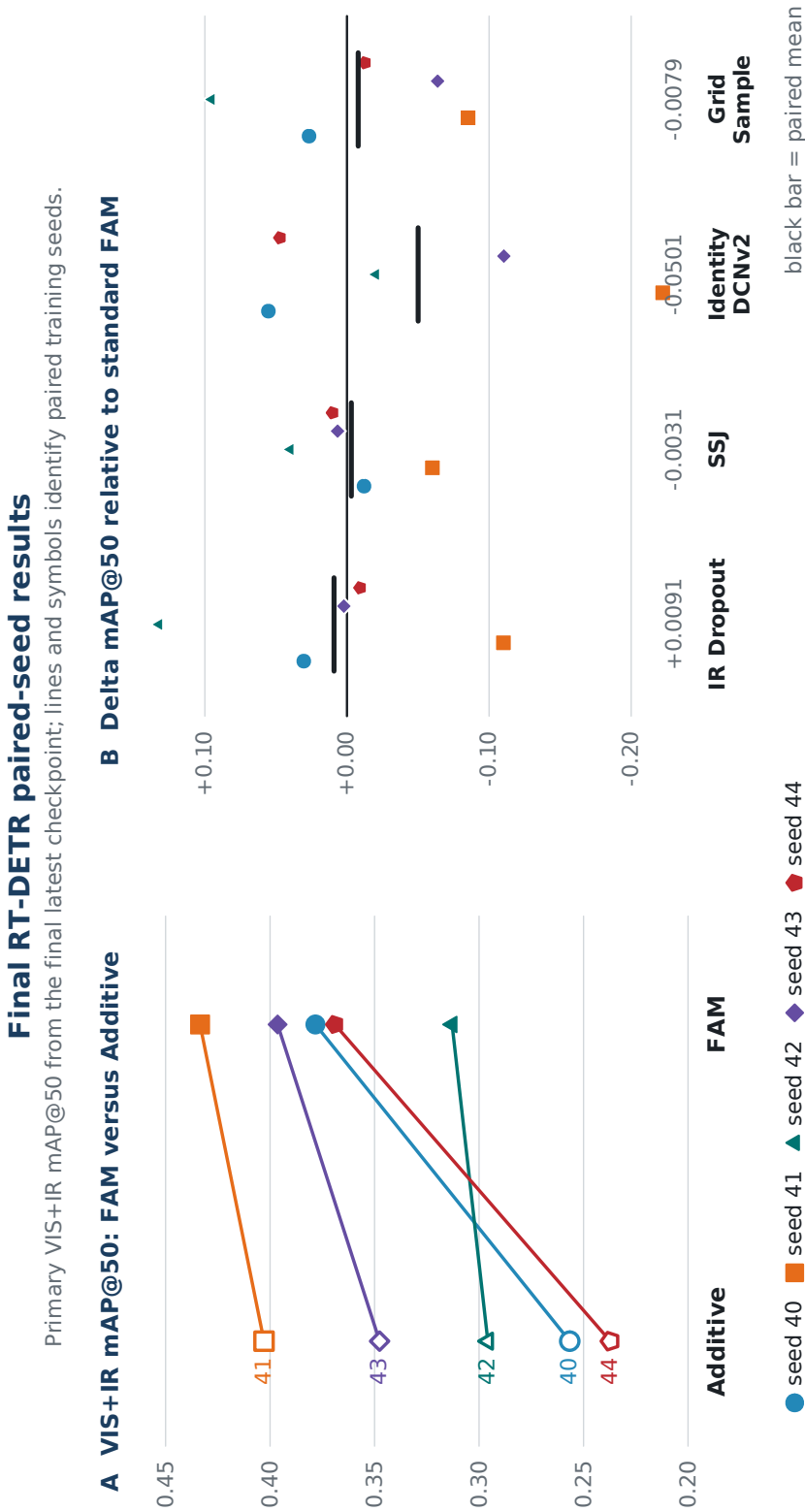


Figure 4.1: Final RT-DETR VIS+IR mAP@50 from the latest checkpoints. Panel A connects Additive and FAM runs initialized with the same training seed; FAM wins all five pairs. Panel B reports, for each seed, the difference between a variant and standard FAM, with black bars denoting paired means. Each seed-specific run/checkpoint is an experimental unit; points are not repeated measurements of one checkpoint.

Table 4.4: Final RT-DETR mAP@50 by input condition, mean $\pm$ sample standard deviation across five checkpoints.

Configuration	VIS	IR	VIS+IR
Additive	0.1543 ± 0.0331	0.1567 ± 0.0464	0.3081 ± 0.0677
FAM	0.2442 ± 0.0307	0.2028 ± 0.0635	0.3780 ± 0.0439
FAM + IR Dropout	0.2368 ± 0.0428	0.2079 ± 0.0267	0.3870 ± 0.0466
FAM + SSJ	0.2312 ± 0.0163	0.1794 ± 0.0416	0.3751 ± 0.0185
Identity DCNv2	0.2150 ± 0.0561	0.1768 ± 0.0686	0.3281 ± 0.0942
Grid Sample	0.2614 ± 0.0267	0.1895 ± 0.0211	0.3701 ± 0.0345

is zero-filled while both backbones and the FAM path remain in the model. The unimodal difference therefore measures the robustness of the complete learned system under missing input, not alignment between two real signals during that inference.

4.2 Mechanistic validation of the FAM

4.2.1 Diagnostic design

The main detection result establishes a predictive advantage but does not identify its mechanism. A frozen diagnostic campaign therefore examined standard FAM, FAM+SSJ, and Grid Sample for seeds 40–44. For every checkpoint, ten predefined samples were drawn from MtErie, ten from the FHL training session, and ten from the Baker training session. Each sample was inspected at P3, P4, and P5:

$$3 \text{ configurations} \times 5 \text{ seeds} \times 30 \text{ samples} \times 3 \text{ levels} = 1,350 \quad (4.1)$$

sample-level observations. They are aggregated within each checkpoint before the five checkpoints are compared.

Forward hooks capture RGB and IR features before FAM, the transformed IR feature, and the predicted offsets and masks. Offset components are converted to input-image pixels by multiplying them by the level stride. Feature maps are projected to three shared PCA components for visualization. These pseudo-colour maps are abstract feature projections, not registered RGB and thermal images.

4.2.2 Offset behavior across levels and seeds

Table 4.5 reports the mean absolute offset component after averaging the thirty samples within each checkpoint. Standard FAM has nonzero transformations at P3 and P4 in every seed. Excluding the pathological P5 entry, offset scale varies both by seed and level.

At P3, SSJ increases the mean offset magnitude in all five paired seeds; the mean difference is $(+1.230 \pm 1.001)$ input pixels. It also reduces the ratio of output-to-input spatial variation in all five checkpoints, producing a smoother transformed feature. P4 offset changes are less consistent, and P5 cannot be paired directly with standard FAM

Table 4.5: Mean absolute offset components in input-image pixels. Values are aggregated over thirty frozen samples per checkpoint.

Configuration	Seed	P3	P4	P5
FAM	40	2.69	3.23	13.69
FAM	41	3.34	7.47	5029.83
FAM	42	3.48	6.32	34.20
FAM	43	3.04	3.74	14.44
FAM	44	5.99	4.80	26.63
FAM + SSJ	40	4.42	9.11	34.45
FAM + SSJ	41	3.45	10.71	23.42
FAM + SSJ	42	5.41	4.73	20.95
FAM + SSJ	43	5.23	6.84	18.02
FAM + SSJ	44	6.17	4.18	36.85
Grid Sample	40	1.65	2.60	25.25
Grid Sample	41	1.70	3.59	7.66
Grid Sample	42	2.08	2.89	11.65
Grid Sample	43	2.29	4.60	15.86
Grid Sample	44	2.41	2.49	10.17

without accounting for the seed-41 collapse. The diagnostic conclusion agrees with the detection result: SSJ changes the learned module in a repeatable way at P3, but that change does not improve final detection accuracy.

Grid Sample has smaller and less dispersed P3/P4 offsets and preserves the input feature more closely. Its same-cell cosine similarity increases slightly after the warp. DCNv2 instead reduces same-cell cosine similarity and increases standardized RMSE because it also filters and mixes channels. This does not show that DCNv2 worsens physical alignment; it shows that its output is not merely the IR tensor shifted to resemble RGB at the same cell.

4.2.3 P5 collapse in seed 41

FAM seed 41 exhibits a complete and repeated P5 degeneration across all thirty samples and all three sessions. Its mean absolute offset is approximately 5,030 input pixels, orders of magnitude larger than the normal checkpoints. The transformed IR output has zero spatial standard deviation and equals the learned DCNv2 bias at every location.

The scale anomaly is already present before FAM: mean P5 global standard deviations are 17.30 for RGB and 20.18 for IR, compared with 0.17–0.69 and 0.056–0.101 in the other seeds. The offset predictor itself has ordinary weight norms. Its enormous offsets are nearly identical across samples (correlation above 0.999) and sample outside the feature map, leaving only the deformable-convolution bias.

The detector nevertheless obtains 0.4335 mAP@50, the highest of the five FAM

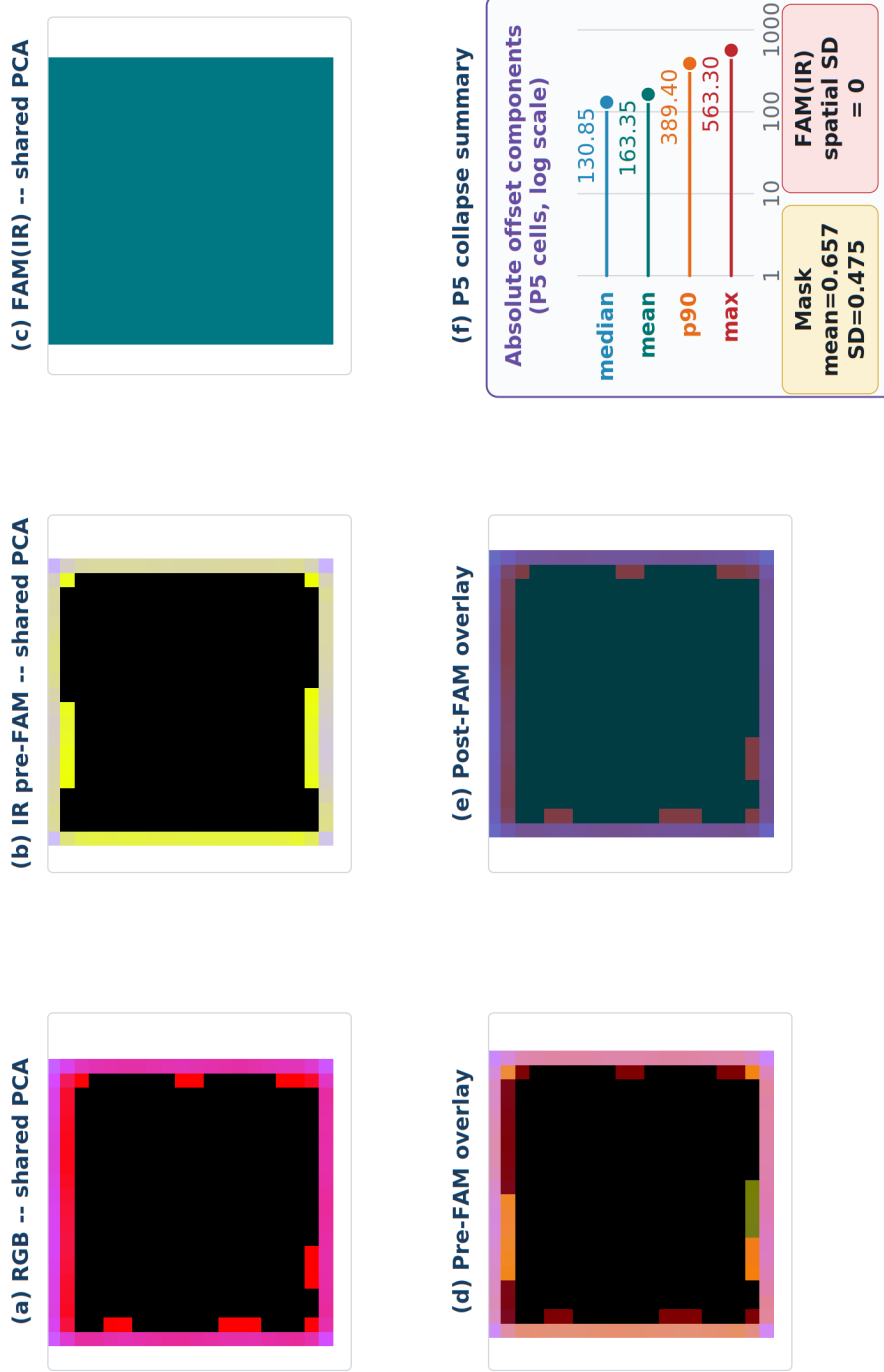


Figure 4.2: Mechanistic diagnostic of the collapsed P5 branch in FAM seed 41 for the prespecified MtErie sample 35. Panels (a)–(c) are pseudo-colour PCA projections obtained with a basis shared across the standardized feature tensors; panels (d)–(e) are qualitative 50% blends and are not tensors consumed by the detector. Panel (f) summarizes absolute DCNV2 offset components in feature-cell coordinates and the modulation-mask statistics. The transformed IR feature is spatially constant, while the mean absolute offset component corresponds to 5,227 input pixels. These quantities diagnose a learned feature-space transformation and must not be interpreted as a camera-registration field.

checkpoints. High output accuracy therefore does not guarantee that every internal alignment level behaves sensibly. The main FAM advantage is not created by the anomaly: after excluding seed 41, FAM still beats Additive in 4/4 paired seeds with mean delta (+0.0799).

4.2.4 Factorial P3/P4/P5 ablation

All eight subsets of active FAM levels were evaluated post-hoc in each of the five trained FAM checkpoints. Bypassing a level replaces its transformed IR feature with the pre-FAM IR feature. The intervention measures sensitivity of the trained system; it does not reproduce training a network with that level absent.

Table 4.6: Post-hoc factorial level ablation. Delta is the paired mean difference from the full trained FAM.

Active FAM levels	Mean mAP@50	SD	Delta from full
None	0.2259	0.0654	-0.1521
P3	0.3223	0.0660	-0.0558
P4	0.2396	0.0983	-0.1384
P5	0.2791	0.0377	-0.0989
P3 + P4	0.3221	0.0701	-0.0559
P3 + P5	0.3648	0.0503	-0.0132
P4 + P5	0.2984	0.0254	-0.0796
P3 + P4 + P5	0.3780	0.0440	0.0000

P3 has the largest and most stable factorial main effect. P5 has a positive mean effect but high variability, dominated by seed 41. Removing P5 from that checkpoint reduces mAP@50 from 0.4335 to 0.2399. The decoder has adapted to, or learned to compensate for, the collapsed representation. Static removal after training is therefore not a repair, and the result does not predict the score of a newly trained P3+P4-only model.

4.2.5 Bounded-offset intervention

The proximal mechanism of the constant output is unbounded sampling far outside the feature map. A single corrective hypothesis was therefore frozen before new training. For every raw offset component, the bounded variant uses

$$\Delta_{\text{eff}} = 4 \tanh\left(\frac{\Delta_{\text{raw}}}{4}\right), \quad (4.2)$$

where the limit is expressed in cells of the current feature level. The derivative at zero is one, and no gate, normalization, residual connection, or backbone modification is introduced. The limit of four cells was chosen from the non-pathological checkpoint distributions before observing the five new training results.

Table 4.7: Automatic VIS+IR mAP@50 for the five bounded-offset runs and the corresponding frozen standard-FAM reference.

Seed	Standard FAM	Bounded FAM	Bounded – standard
40	0.3783	0.4208	+0.0426
41	0.4335	0.3319	-0.1015
42	0.3129	0.4214	+0.1085
43	0.3964	0.2735	-0.1229
44	0.3690	0.2998	-0.0692
Mean $\pm$ SD	0.3780 $\pm$ 0.0440	0.3495 $\pm$ 0.0686	-0.0285

The bounded model wins in only 2/5 seeds and has greater dispersion. Its standalone modality evaluation obtains 0.2252 ± 0.0544 VIS, 0.1943 ± 0.0272 IR, and 0.3496 ± 0.0685 VIS+IR, all below the standard-FAM means.

The internal intervention nevertheless succeeds. Every observed effective offset remains below four cells, including a maximum of 3.9915 at P5 seed 41, and none of the 150 inspected P5 checkpoint-sample cases has a spatially constant output. Raw offsets would still reach 13.69 cells, showing that the bound is active. The experiment therefore separates two outcomes: the targeted failure mode is removed, but detection does not improve. Standard FAM remains the selected model.

4.3 Detection error analysis

4.3.1 Protocol

The error analysis compares the five Additive and five standard-FAM final checkpoints on all 708 MtErie frames and 1,770 objects. Predictions are matched one-to-one to annotations at IoU at least 0.50. Threshold 0.01, already used to collect predictions for the final evaluations, is the primary collection threshold; 0.05, 0.10, 0.25, and 0.50 form a predefined sensitivity analysis. The analysis does not select a deployment threshold.

Of the 1,770 objects, 1,451 are COCO-small, 301 medium, and 18 large at the evaluation resolution. Large-object results are therefore particularly uncertain.

4.3.2 Results across thresholds

Recall is the most consistent difference. FAM is better in 5/5 seeds at thresholds 0.10, 0.25, and 0.50. At 0.25, recall increases from 0.2340 to 0.3330, a mean paired gain of 0.0990, and the fraction of non-empty frames with at least one missed target decreases by 0.0824.

The gain is not limited to the 18 large objects. At threshold 0.25, mean recall increases by 0.0815 for small and 0.1781 for medium targets. False positives do not improve uniformly: FAM produces slightly fewer on average at low thresholds and slightly more at 0.25 and 0.50, with signs that vary across seeds. The thesis therefore

Table 4.8: Paired RT-DETR error behavior across confidence thresholds. Values are mean $\pm$ sample standard deviation over five checkpoints.

Threshold	Model	Precision	Recall	FP/image	Non-empty frames with FN
0.01	Additive	0.0418 ± 0.0175	0.7677 ± 0.0490	51.006 ± 21.591	0.4508 ± 0.0777
0.01	FAM	0.0443 ± 0.0132	0.8044 ± 0.0328	47.623 ± 17.873	0.4009 ± 0.0524
0.05	Additive	0.2713 ± 0.0909	0.5932 ± 0.1134	4.844 ± 3.199	0.6476 ± 0.1155
0.05	FAM	0.3010 ± 0.0940	0.6539 ± 0.0972	4.336 ± 2.191	0.5835 ± 0.0775
0.10	Additive	0.4721 ± 0.0923	0.4582 ± 0.1535	1.513 ± 1.115	0.7466 ± 0.1223
0.10	FAM	0.5194 ± 0.1148	0.5507 ± 0.1333	1.447 ± 0.792	0.6554 ± 0.0725
0.25	Additive	0.7290 ± 0.0866	0.2340 ± 0.1468	0.249 ± 0.218	0.8682 ± 0.0879
0.25	FAM	0.7715 ± 0.0742	0.3330 ± 0.1506	0.276 ± 0.168	0.7858 ± 0.0525
0.50	Additive	0.9140 ± 0.0536	0.1088 ± 0.0705	0.029 ± 0.033	0.9257 ± 0.0588
0.50	FAM	0.9350 ± 0.0504	0.1627 ± 0.0663	0.036 ± 0.037	0.8790 ± 0.0359

does not claim that FAM reduces false positives.

At threshold 0.01, both configurations generate approximately 48–51 false positives per image. This threshold is appropriate for collecting a broad set of candidates for AP computation but is not an operational alarm threshold.

4.3.3 Qualitative examples

The qualitative manifest was built from ground-truth annotations before inference. For each of the three MtErie sequences, it selects one frame with a small target and one empty frame. Seed 43 is used because its FAM–Additive mAP@50 difference equals the median of the five paired differences; it is neither the best nor worst pair.

The corresponding threshold-0.01 montage is retained in the appendix because the large number of low-confidence boxes makes it difficult to read. In the three predefined small-target frames at 0.25, FAM recovers one target missed by Additive in one frame; both models miss the targets in the other two. No general conclusion is drawn from these six examples.

4.4 Carnation scale-mismatch stress test

4.4.1 Frozen design

Carnation contains a paired visual and thermal acquisition with a substantially stronger scale mismatch than the main splits. Its protocol was frozen before inference and admits only the five final Additive and five final FAM checkpoints. The intersection of frame identifiers contains 739 examples. Each checkpoint is evaluated in VIS+IR, VIS, and IR, yielding thirty evaluations without training, tuning, checkpoint selection, or threshold selection.

Carnation is a deliberately difficult single acquisition. It is not a representative sample of SAR domains, a new validation set, or an estimate of general domain generalization.

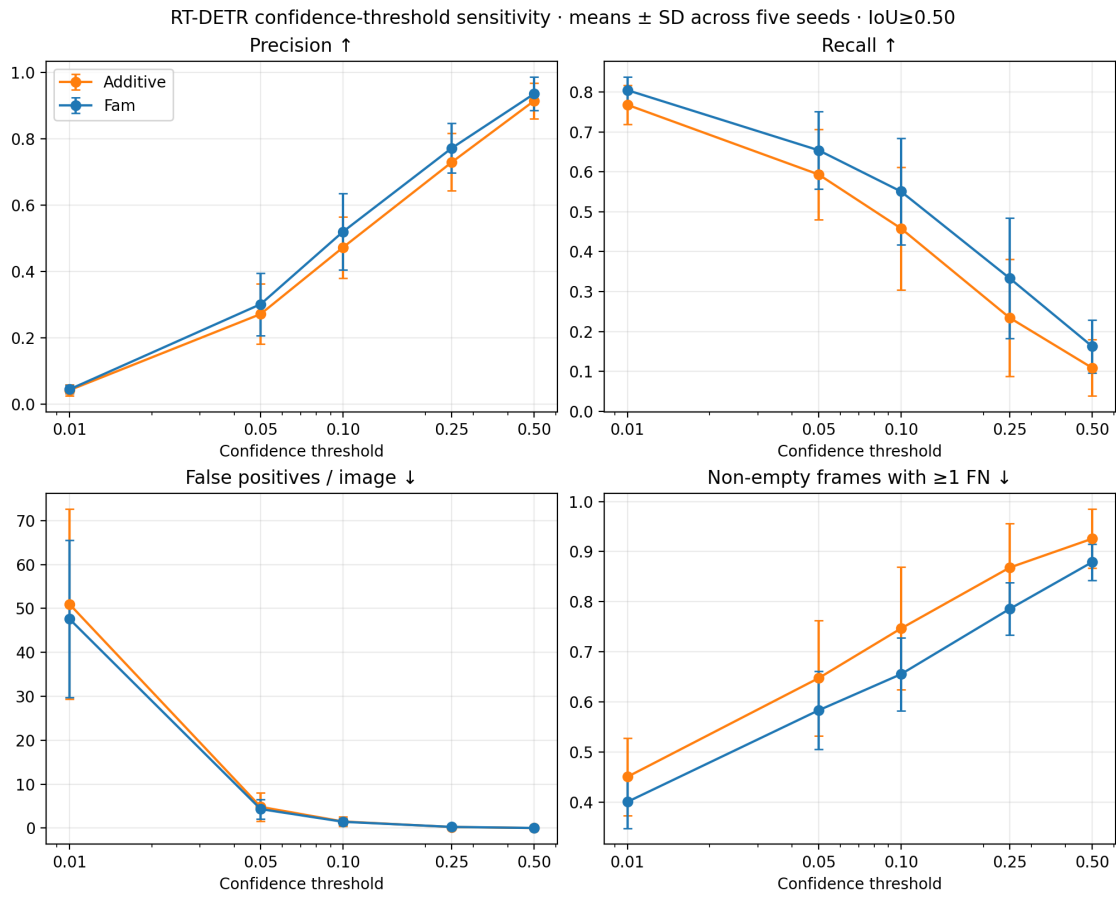


Figure 4.3: Sensitivity of precision, recall, false positives per image, and non-empty frames with a false negative to the five predefined confidence thresholds. Error bars are sample standard deviations across checkpoints.

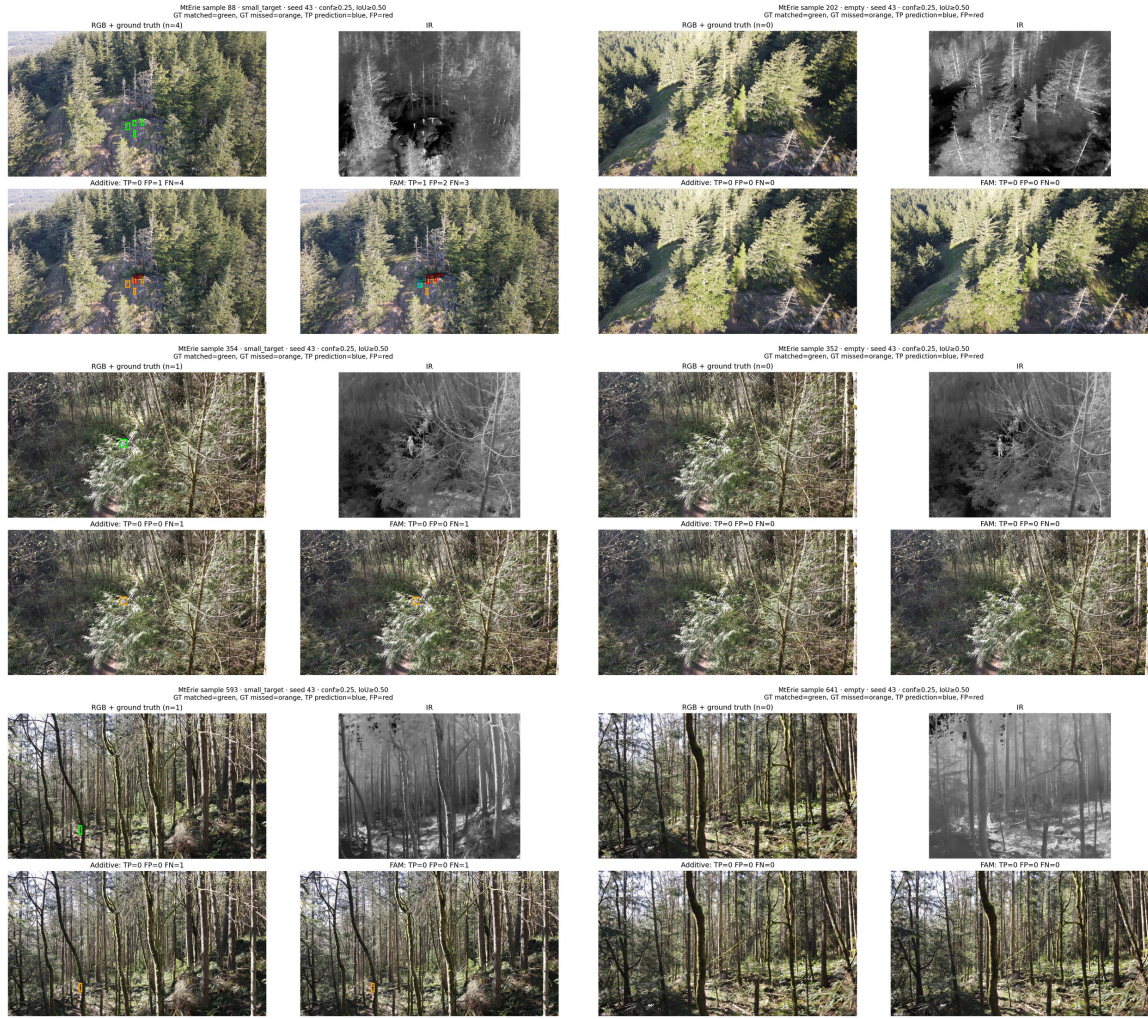


Figure 4.4: Predefined qualitative examples at confidence 0.25. Each row contains one small-target frame and one empty frame from a MtErie sequence. The same frames and display rules are used for Additive and FAM. These examples illustrate, but do not replace, the aggregate analysis.

4.4.2 Results

Table 4.9: Carnation mAP@50 across five frozen checkpoints.

Input	Additive		FAM	Mean paired delta	FAM wins
VIS+IR	0.1445 $\pm$ 0.0709	0.2408 $\pm$ 0.0380		+0.0962	4/5
VIS	0.1555 $\pm$ 0.0669	0.3196 $\pm$ 0.0518		+0.1641	5/5
IR	0.1288 $\pm$ 0.0776	0.0979 $\pm$ 0.0652		-0.0308	1/5

FAM is more tolerant than Additive in the fusion condition, with a mean gain of 0.0962 and four paired wins. This is not sufficient to make fusion the best input condition. For the FAM checkpoints, VIS-only exceeds VIS+IR in all five seeds by a mean of 0.0789. Adding the strongly mismatched IR signal is therefore harmful relative to the available visible stream even after FAM transformation.

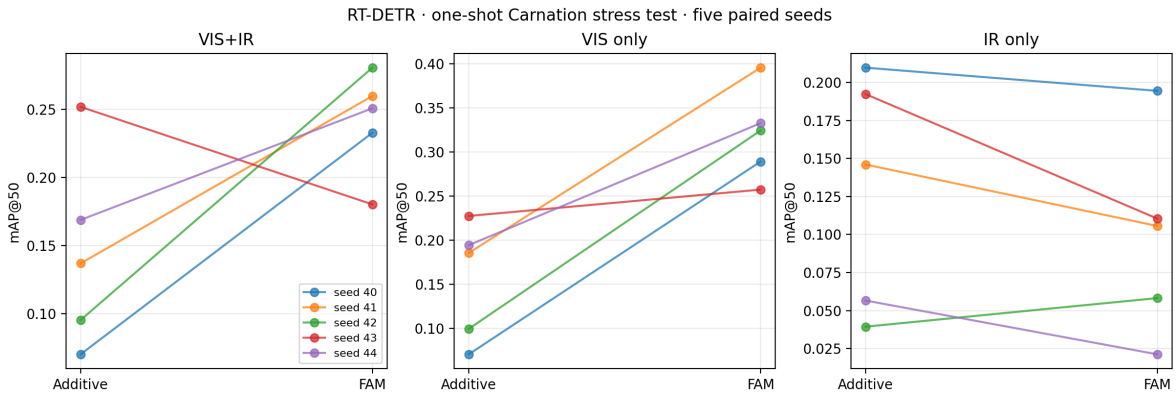


Figure 4.5: Paired Carnation results for the three modality conditions. The stress test shows partial damage mitigation by FAM, not correction of the severe mismatch.

The result also bounds the internal MtErie finding that VIS+IR was always better than the strongest single modality. That conclusion is stable within the main benchmark but does not survive this extreme acquisition.

4.5 Computational cost

The compute comparison uses the seed-43 Additive and FAM checkpoints, chosen in advance as the same typical pair used for qualitative illustrations. Each configuration is measured in three isolated processes on an NVIDIA GeForce RTX 4070 Laptop GPU with PyTorch 2.4.0, CUDA 12.1, batch size one, FP32 input of 640×640 , thirty warm-up iterations, and one hundred timed forward passes per trial. TF32, autocast, preprocessing, and postprocessing are excluded.

Relative to Additive, FAM adds 77.5% parameters, 46.2% proxy computation, 17.9% detector latency, and 275.92 MiB of total peak CUDA memory. The proxy adds an analytical dense-convolution equivalent for DCNv2 because the profiler does not count that operator fully; it still omits parts of bilinear sampling, mask application, and

Table 4.10: Detector-only computational benchmark. The operation count is a proxy and the latency uncertainty is the sample standard deviation across three trial means.

Model	Parameters	Proxy GFLOPs	Latency	Throughput	CUDA peak
Additive	66.20 M	208.37	66.91 ± 0.30 ms	14.95 image/s	436.26 MiB
FAM	117.49 M	304.54	78.87 ± 0.37 ms	12.68 image/s	712.18 MiB

sigmoid evaluation. Measured latency is therefore the more operational quantity, while remaining specific to this GPU and detector scope.

The FAM improvement is not computationally free. Deployment decisions must weigh its recall and mAP@50 gains against memory, throughput, energy, and the un-measured cost of the full UAV pipeline.

4.6 Architectural transferability to YOLOv10

4.6.1 Dual-backbone architecture

The YOLOv10-s transfer model receives a four-channel paired tensor and splits it into a three-channel RGB stream and one-channel IR stream. Two independent backbones produce P3, P4, and P5 features at strides 8, 16, and 32. The Additive configuration sums each pair; the FAM configuration applies the same standard DCNv2 module to IR before addition. The resulting features enter the shared YOLOv10 FPN/PAN neck and detection head.

Missing modalities are implemented through feature gating. During training, one branch or both are selected with probabilities 0.2/0.2/0.6. In standalone VIS or IR evaluation, the excluded backbone is not executed and FAM is bypassed because a paired reference is unavailable. Unimodal differences therefore reflect how training with FAM shaped the remaining weights, not a direct alignment operation at inference.

4.6.2 Exploratory history and checkpoint problem

Initial YOLO experiments used single runs, `best.pt`, and variable early stopping selected through the small FHL validation set. They explored fusion-only training, input-level modality masking, feature gating, FAM, and SSJ. These runs helped identify feature-level gating as the appropriate missing-modality contract but do not provide a final transfer claim.

The historical seed-42 Additive and FAM checkpoints were selected at epochs 16 and 89 and achieved higher MtErie values than the final 200-epoch checkpoints. Because FHL is small and strongly shifted, its curve does not define a stable general rule. Selecting 100 epochs, reusing `best.pt`, or choosing a new epoch after observing MtErie would be retrospective test tuning.

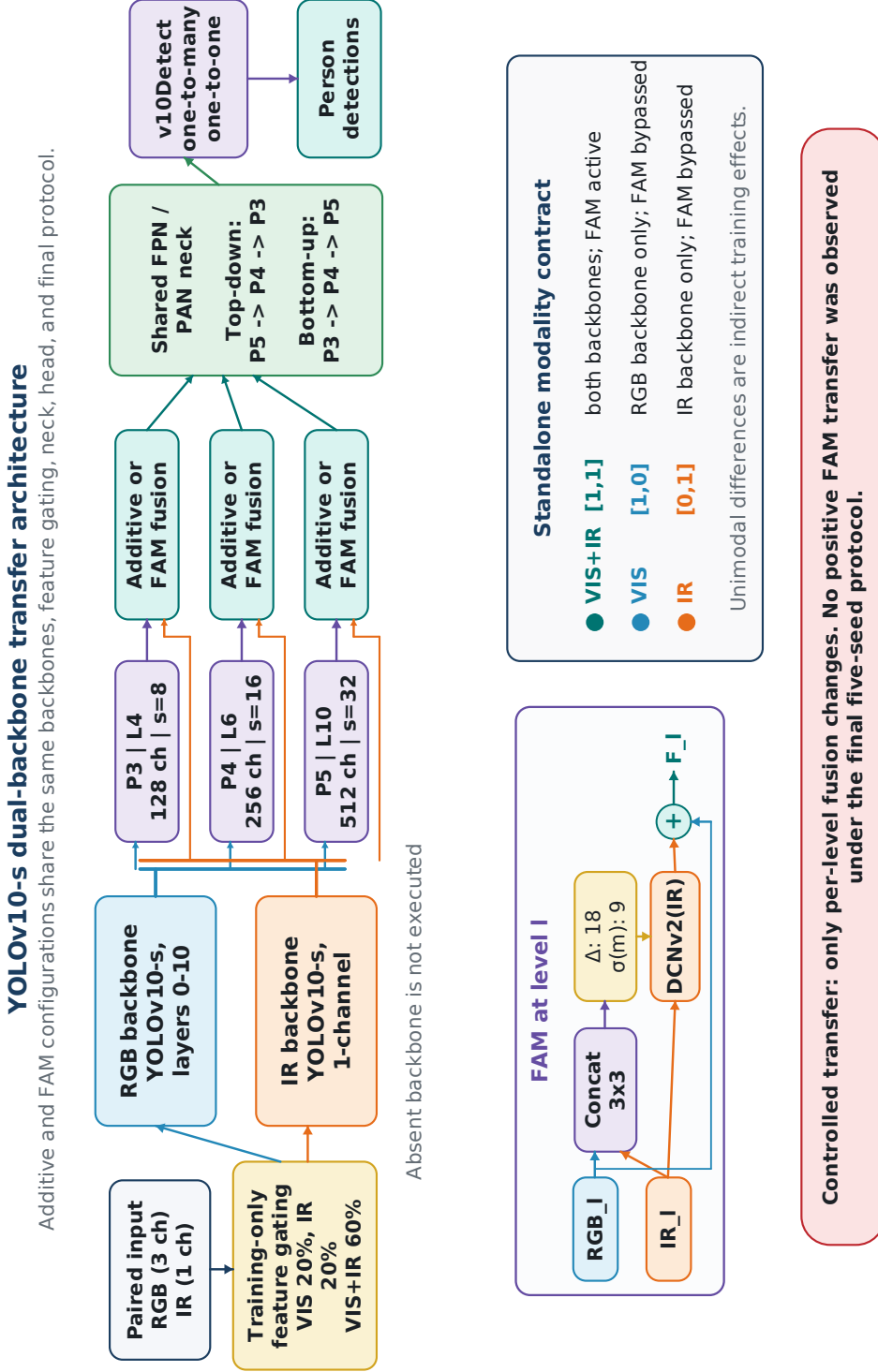


Figure 4.6: Implemented YOLOv10-s transfer architecture. The controlled comparison changes the P3/P4/P5 fusion operation while retaining the dual backbones, feature gating, FPN/PAN neck, detection head, and final protocol. During unimodal inference the excluded backbone is not executed and FAM is bypassed; consequently, unimodal Additive-FAM differences are indirect effects of training rather than direct alignment effects.

4.6.3 Final YOLOv10 protocol

The final comparison fixes two configurations: dual-backbone Additive and dual-backbone standard FAM, both with feature gating 20/20/60. Each uses paired seeds 40–44, an isolated Python process, 200 fixed epochs, disabled early stopping, no validation-based selection, and the final `last.pt` checkpoint. The 200-epoch horizon was retained because it was declared in the historical YAML schedule before the final results were observed.

All ten runs completed 200 epochs. Their automatic VIS+IR results are shown in Table 4.11.

Table 4.11: Final automatic YOLOv10 VIS+IR mAP@50 at `last.pt`.

Seed	Additive	FAM	FAM – Additive
40	0.2304	0.1883	-0.0421
41	0.2740	0.2749	+0.0009
42	0.2494	0.1947	-0.0546
43	0.2159	0.2603	+0.0444
44	0.2728	0.1802	-0.0926
Mean $\pm$ SD	0.2485 $\pm$ 0.0256	0.2197 $\pm$ 0.0443	-0.0288 $\pm$ 0.0528

FAM wins in 2/5 seeds. The paired 95% t -interval is $[-0.0944, +0.0368]$, with exploratory paired t -test $p = 0.2894$ and exact Wilcoxon $p = 0.4375$. The final protocol does not support a YOLOv10 fusion benefit.

4.6.4 Standalone modality evaluation

The same standalone evaluator was used for all three input conditions on the ten final checkpoints. Its results are in Table 4.12.

Table 4.12: Final standalone YOLOv10 mAP@50 by modality, mean $\pm$ SD over five checkpoints.

Input	Additive	FAM	Mean delta	FAM wins
VIS+IR	0.2498 $\pm$ 0.0254	0.2213 $\pm$ 0.0441	-0.0285	2/5
VIS	0.1964 $\pm$ 0.0218	0.1929 $\pm$ 0.0407	-0.0035	2/5
IR	0.0277 $\pm$ 0.0032	0.0369 $\pm$ 0.0009	+0.0092	5/5

The standalone fusion values reproduce the automatic ranking. VIS shows no stable benefit. IR improves in all seeds but remains too low to be operationally useful, and FAM is bypassed in that condition. The difference is an indirect training effect on the learned backbones and neck.

The standalone evaluator produced VIS+IR AP values systematically about 0.0010–0.0021 above the automatic Ultralytics validator. The maximum difference exceeded the predefined 0.002 sanity tolerance by 0.000077. The tolerance was not changed.

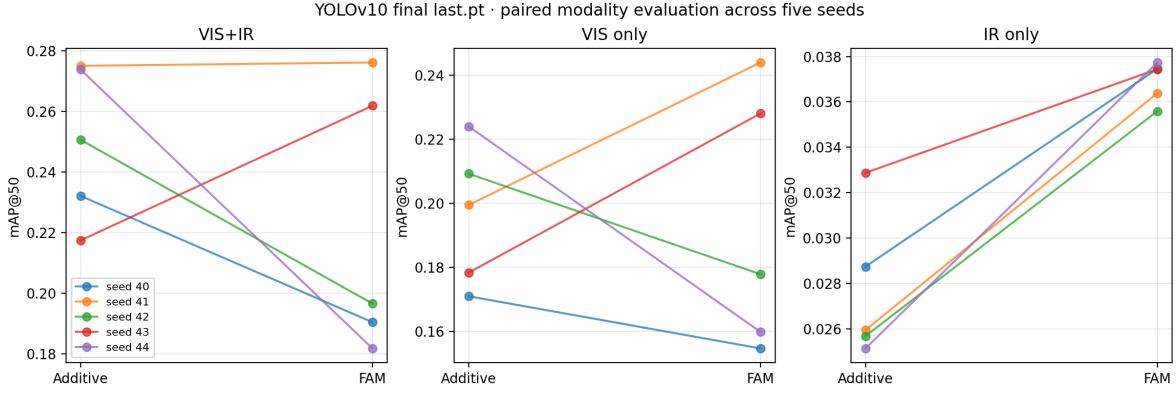


Figure 4.7: Final YOLOv10 paired modality evaluation. The favorable but very small IR-only delta does not compensate for the negative fusion result.

Since all standalone modality conditions use the same evaluator and the checkpoint ranking is preserved, standalone values are used for the modality comparison while the automatic table remains the primary training result.

4.6.5 Training-horizon interpretation

At seed 42, the historical early Additive checkpoint obtained 0.3469 at epoch 16 and the historical FAM checkpoint 0.2965 at epoch 89, compared with 0.2494 and 0.1947 at epoch 200. Additive training losses and learning rate match the old run exactly until its early stop, after which the final run continues. Training losses keep decreasing while the final test score is lower than the historical early score.

This pattern is compatible with late overfitting or specialization to the training sessions, but training loss alone does not establish the cause. The valid conclusion is that the declared 200-epoch protocol yields a negative FAM transfer result. A new YOLO study would first require a representative validation set constructed only from training sessions and a checkpoint rule fixed before observing MtErie.

4.7 Cross-architecture summary

The three extension families provide different strengths of evidence. Deformable DETR shows an exploratory increase in median VIS+IR from 0.249 to 0.294 when FAM is added, but its intervals overlap and its protocol predates the final controls. Complete DINO base is weak under the studied configuration, so FAM is not tested on that implementation. YOLOv10 uses a final paired-seed protocol and shows a negative mean FAM effect.

The thesis therefore does not demonstrate positive architectural transferability. It demonstrates that the FAM concept can be implemented in multiple detector families, that a promising exploratory trend exists in one, and that the controlled YOLOv10 transfer does not reproduce the RT-DETR gain.

Chapter 5

Discussion and Conclusions

5.1 Answers to the research questions

5.1.1 RQ1: experimental reliability

The observed RT-DETR variability is large enough to change single-run model rankings. Nominally identical native-CUDA executions diverge after the first backward pass even when initialization, input, and initial loss agree. The transferred COCO person head reduces amplification of those perturbations but does not remove them, and DCNv2 prevents strict deterministic FAM training in the studied environment.

Reliable comparison consequently requires more than recording a seed. The final protocol combines paired seed values, isolated processes, a fixed training horizon, final checkpoints, identical pretrained-head transfer, training-only Modal Dropout, and distributional reporting. The validation audit is equally important: a small shifted validation set can be informative about domain mismatch while remaining unsuitable for selecting a checkpoint representative of MtErie.

The methodological answer to RQ1 is therefore that experimental design changes the scientific conclusion. The move from historical single runs to paired replication replaces the claim that SSJ is the best variant with a more limited and repeatable selection of standard FAM.

5.1.2 RQ2: RT-DETR effectiveness and missing modalities

On the locked MtErie benchmark, standard FAM improves Additive fusion in every paired seed and raises mean mAP@50 by 0.0700. The improvement is supported by COCO mAP, mAP@75, mAR, and the later recall analysis. IR Dropout and SSJ do not provide a reliable mean gain over standard FAM; Identity DCNv2 is highly variable, and Grid Sample is close but not consistently superior.

VIS+IR is better than the strongest single modality for all thirty final RT-DETR checkpoints. Modal Dropout therefore supports a usable response when a modality is removed without making the detector indifferent to the availability of both signals. On the internal benchmark, multimodal information remains valuable.

The answer is scoped to the internal development benchmark. It establishes a consistent paired advantage for FAM over Additive under the selected protocol, not a

universal expected gain for all seeds, datasets, sensor geometries, or implementations.

5.1.3 RQ3: mechanism, levels, and bounded offsets

The FAM predicts nonzero transformations at P3 and P4 across all five standard checkpoints. Its offset scale and transformed features vary with seed, level, and acquisition session. SSJ changes P3 behavior consistently, increasing offset magnitude and smoothing, even though detection does not improve. Grid Sample is internally more stable but does not establish that a pure warp is equivalent to DCNv2.

The diagnostics do not demonstrate physical RGB-IR registration. DCNv2 mixes geometric sampling, filtering, mask modulation, and channel interaction. Its output becomes less similar to RGB at the same cell under the selected feature proxies, while still improving detection. The supported interpretation is a task-oriented geometric-functional transformation.

The P5 seed-41 collapse is an important counterexample to reasoning from output accuracy alone. An internally constant branch coexists with the best FAM checkpoint because the rest of the detector compensates for it. Factorial ablation identifies P3 as the most stable contributor and P5 as positive but variable; removing P5 after training is especially damaging in the collapsed checkpoint.

Smoothly bounding offsets at four feature cells eliminates the observed out-of-map collapse in five new runs, but lowers mean accuracy and increases dispersion. The answer to the RQ3 sub-question is therefore asymmetric: the intervention is mechanistically effective but is not a better detector.

5.1.4 RQ4: operational relevance

FAM primarily changes the recall side of the precision-recall trade-off. At confidence 0.25, it improves mean recall by 0.0990 in all five paired seeds and reduces the frequency of non-empty frames with at least one missed target. This is relevant to SAR, where a missed person can have high cost. False positives do not decrease consistently, however, and the 0.01 threshold used for AP collection is not operational because it produces roughly fifty false positives per image.

Carnation exposes a geometric boundary. FAM is more tolerant than Additive fusion under the severe scale mismatch, but adding IR remains worse than using VIS alone in every FAM checkpoint. FAM mitigates the damage of that mismatch; it does not solve arbitrary scale differences between sensors.

The performance gain also has a material computational cost: 77.5% more parameters, 46.2% more proxy GFLOPs, 17.9% more detector latency, and higher CUDA memory on the measured GPU. These values prevent describing FAM as a free or generically real-time improvement.

5.1.5 RQ5: architectural transferability

Positive transfer is not established. Deformable DETR provides a promising exploratory median trend, but its uncertainty intervals overlap and its protocol is not aligned with the final campaign. Complete DINO base is too weak under the studied configuration

to justify FAM variants. Under the controlled YOLOv10 protocol, FAM decreases mean VIS+IR performance and wins in only two of five seeds.

The negative YOLOv10 result is scientifically useful. It shows that the RT-DETR gain depends on the surrounding representation, fusion interface, optimization trajectory, or checkpoint regime and cannot be assumed from the local FAM operation alone. It does not show that YOLOv10 or DINO are intrinsically unsuitable for RGB-thermal detection.

5.2 Overall interpretation

The most defensible conclusion is narrower than the historical narrative but better supported. Standard FAM is the strongest studied RT-DETR configuration because it improves Additive consistently on the internal benchmark, not because it achieves the largest single score or because its offsets reproduce a known camera transform. Its main observed detection benefit is higher recall.

At the same time, the module is internally variable and computationally expensive. One feature level can collapse without visible degradation in the headline metric, and a correction that makes the mechanism safer can make the detector worse. Architecture transfer and external scale mismatch further show that an alignment-inspired inductive bias is not sufficient on its own.

These results support a broader methodological point: evaluating multimodal fusion should combine output metrics, missing-modality tests, paired replication, internal diagnostics, out-of-distribution stress, and cost. Any one of these views alone would have produced an incomplete model selection.

5.3 Threats to validity

5.3.1 Internal validity

Native CUDA execution is nondeterministic, particularly for deformable operators. Pairing seeds controls intended stochastic sources but cannot pair the microscopic CUDA perturbations of two architectures. Five independent checkpoints estimate the resulting distribution but do not exhaust it.

The level ablations are interventions on already trained networks. Bypassed representations are outside the training path and cannot be interpreted as the performance of a newly trained P3-only or P3+P5 architecture. The P5 audit and the choice of a bounded correction followed observation of the MtErie checkpoint. Freezing one hypothesis before five new trainings reduces further adaptation, but the correction remains post-hoc with respect to the internal benchmark.

The standalone YOLOv10 evaluator differs slightly from the automatic Ultralytics validator. The predefined tolerance is recorded as missed rather than changed after inference. Comparisons among modalities use the same standalone evaluator, while automatic VIS+IR results remain primary for the training protocol.

5.3.2 Construct validity

mAP@50 emphasizes detection at IoU 0.50 and is not a complete measure of localization quality or SAR utility. Secondary COCO mAP, mAP@75, mAR, and error metrics reduce dependence on one score but do not measure response time, operator workload, alarm calibration, energy consumption, or consequences of false alarms.

Offset magnitude, feature cosine similarity, RMSE, smoothing, and PCA are internal proxies. Without camera calibration or ground-truth correspondences, they do not validate physical registration. PCA figures also depend on a low-dimensional projection and are used illustratively.

5.3.3 Statistical conclusion validity

Five paired runs permit estimation of effect direction and variability but produce wide intervals. Exact Wilcoxon tests have coarse resolution, and the smallest attainable two-sided value for five nonzero signed pairs is 0.0625. The reported tests are exploratory and are not corrected for the number of contrasts. Checkpoints, rather than frames, boxes, diagnostic samples, or feature cells, are the experimental units.

5.3.4 External validity

MtErie was repeatedly consulted during project development and is not a blind holdout. Its final locked comparisons are internally useful but may overstate performance on a new acquisition. The FHL validation split is too small and shifted to support ordinary checkpoint selection for MtErie.

Carnation is one deliberately severe scale-mismatch case. It reveals a specific boundary but cannot estimate performance across SAR domains. The study uses one dataset, one main RT-DETR size, one YOLOv10 size, and one GPU for latency. Different cameras, calibration, target scales, thermal conditions, hardware, or training budgets may change the conclusions.

Deformable DETR and DINO results use the previous protocol. They cannot form a quantitatively homogeneous architecture leaderboard with final RT-DETR and YOLOv10.

5.4 Future work

5.4.1 A representative model-selection protocol

The first priority is not another grid on MtErie. A future study should build a validation split exclusively from training data, separated by acquisition session or temporal block to reduce leakage. The checkpoint rule, primary metric, training horizon, and allowed hyperparameter search should be fixed before the internal test or a new holdout is observed. A genuinely independent paired acquisition would then be reserved for final confirmation.

This protocol is necessary before deciding whether the late YOLOv10 degradation is caused by overfitting, an unsuitable schedule, or validation mismatch. It would also

allow early stopping to be compared fairly across architectures.

5.4.2 Improving the alignment architecture

The diagnostics suggest several targeted directions for a future improved model:

1. normalize or condition the offset predictor on feature scale so that a backbone activation anomaly is less likely to generate extreme offsets;
2. add a residual identity path or a learned per-level gate so that raw IR information remains available if the transformed branch becomes unreliable;
3. regularize offset magnitude smoothly rather than applying a hard performance-independent bound;
4. train level-specific variants, particularly P3-focused or gated-P5 models, instead of inferring their behavior from post-hoc bypasses;
5. use cross-modal consistency or modality-specific auxiliary losses to improve the weak branch without requiring the fusion output to dominate every training step;
6. exploit calibration metadata or synthetic known transformations to introduce geometric supervision and distinguish registration from task-oriented feature transformation.

These candidates should not all be searched on MtErie. A small number of mechanistically justified variants should be frozen using the new validation protocol, followed by paired replication and the same internal audit used in this thesis.

5.4.3 Broader evaluation and deployment

Further work should include multiple independent RGB–thermal acquisitions, controlled scale and translation perturbations, additional missing-sensor patterns, confidence calibration, and end-to-end latency including image loading, preprocessing, transfer, postprocessing, and visualization. Deployment experiments should report power and memory limits of the target UAV hardware rather than extrapolating from a laptop GPU.

5.5 Conclusions

This thesis evaluated RGB–thermal fusion for UAV Search and Rescue from a reliability-first perspective. It began with an inherited RT-DETR Feature Alignment Module and replaced a single-run architectural narrative with a paired multi-seed comparison, direct inspection of learned transformations, level interventions, external stress testing, and transfer analysis.

Standard FAM improves Additive RT-DETR on the internal MtErie benchmark in all five paired seeds and consistently increases recall at practical sensitivity thresholds. It remains the selected RT-DETR model. The internal evidence, however, supports a

learned geometric-functional transformation rather than verified physical registration. The P5 collapse, the negative bounded-offset result, the Carnation VIS-only advantage, the computational overhead, and the negative YOLOv10 transfer collectively define where the improvement stops.

The principal outcome is therefore both architectural and methodological: FAM is promising within RT-DETR, but a multimodal detector should not be judged by its best checkpoint or its fusion score alone. Repeated runs, explicit missing-modality evaluation, internal diagnostics, external stress cases, and computational cost are necessary to determine whether an apparent fusion gain is reliable, understandable, and transferable.

Appendix A

Supplementary Results and Artifact Provenance

A.1 Secondary RT-DETR metrics

Table A.1: Final VIS+IR secondary metrics, mean $\pm$ sample standard deviation over five training checkpoints. COCO mAP averages IoU thresholds from 0.50 to 0.95.

Configuration	COCO mAP	mAP@50	mAP@75	mAR@100
Additive	0.1059 $\pm$ 0.0243	0.3080 $\pm$ 0.0677	0.0474 $\pm$ 0.0128	0.2474 $\pm$ 0.0302
FAM	0.1351 $\pm$ 0.0096	0.3780 $\pm$ 0.0440	0.0692 $\pm$ 0.0043	0.2798 $\pm$ 0.0141
FAM + IR Dropout	0.1360 $\pm$ 0.0165	0.3871 $\pm$ 0.0469	0.0698 $\pm$ 0.0079	0.2764 $\pm$ 0.0222
FAM + SSJ	0.1297 $\pm$ 0.0095	0.3749 $\pm$ 0.0185	0.0625 $\pm$ 0.0086	0.2719 $\pm$ 0.0331
Identity DCNv2	0.1134 $\pm$ 0.0305	0.3280 $\pm$ 0.0943	0.0560 $\pm$ 0.0141	0.2556 $\pm$ 0.0357
Grid Sample	0.1331 $\pm$ 0.0077	0.3701 $\pm$ 0.0345	0.0631 $\pm$ 0.0044	0.2849 $\pm$ 0.0115

A.2 Low-threshold qualitative montage

A.3 Versioned compact artifacts

Frame-level prediction outputs and diagnostic JSON files are retained in the local `out/` directories. The following compact artifacts are stored with the thesis source:

Artifact	Content
<code>results/rtdetr_error_analysis_checkpoints.csv</code>	Fifty checkpoint-threshold summaries for Additive and FAM on MtErie.
<code>results/rtdetr_carnation_stress_test.csv</code>	Thirty checkpoint-modality units from the frozen Carnation stress test.
<code>results/rtdetr_compute_benchmark.csv</code>	Parameter, operation-proxy, latency, throughput, and CUDA-memory summary.

Artifact	Content
results/yolo_final_modality_evaluation.csv	Thirty final YOLOv10 checkpoint–modality evaluations.
images/rtdetr_additive_fam_error_summary.png	Paired error summary at confidence 0.01.
images/rtdetr_additive_fam_threshold_sensitivity.png	Multi-threshold error sensitivity analysis.
images/rtdetr_carnation_paired_map50.png	Carnation paired-seed plot.
images/yolov10_final_modality_paired_map50.png	Final YOLOv10 paired modality plot.
images/experimental_timeline.pdf	Vector chronology separating inherited, exploratory, audit, and locked-protocol stages.
images/rtdetr_fam_architecture.pdf	Vector diagram of the implemented dual-backbone RT-DETR and per-level FAM.
images/rtdetr_final_paired_map50.pdf	Final paired-seed RT-DETR comparison generated from the primary training outputs.
images/fam_p5_seed41_collapse.png	Thesis-readable P5 collapse diagnostic assembled from the saved feature diagnostic and JSON statistics.
images/yolov10_dual_backbone_fam.pdf	Vector diagram of the implemented YOLOv10-s transfer architecture and modality contract.

The artifact README records protocol identifiers, completeness markers, and checksums of the larger local aggregates. Raw project reports are retained as historical sources, while the updated Markdown audits remain the authoritative source for final conclusions.

A.4 Code availability

Model implementations, preprocessing code, experimental configurations, and analysis scripts are maintained in the project repository. The five thesis figures introduced above can be regenerated with `scripts/generate_thesis_figures.py`. Historical public project material is available at https://github.com/DonatoFe11/SAR_Fusion. Checkpoints required for the reported final diagnostics are preserved locally because model-weight files were excluded from the experiment tracker uploads.

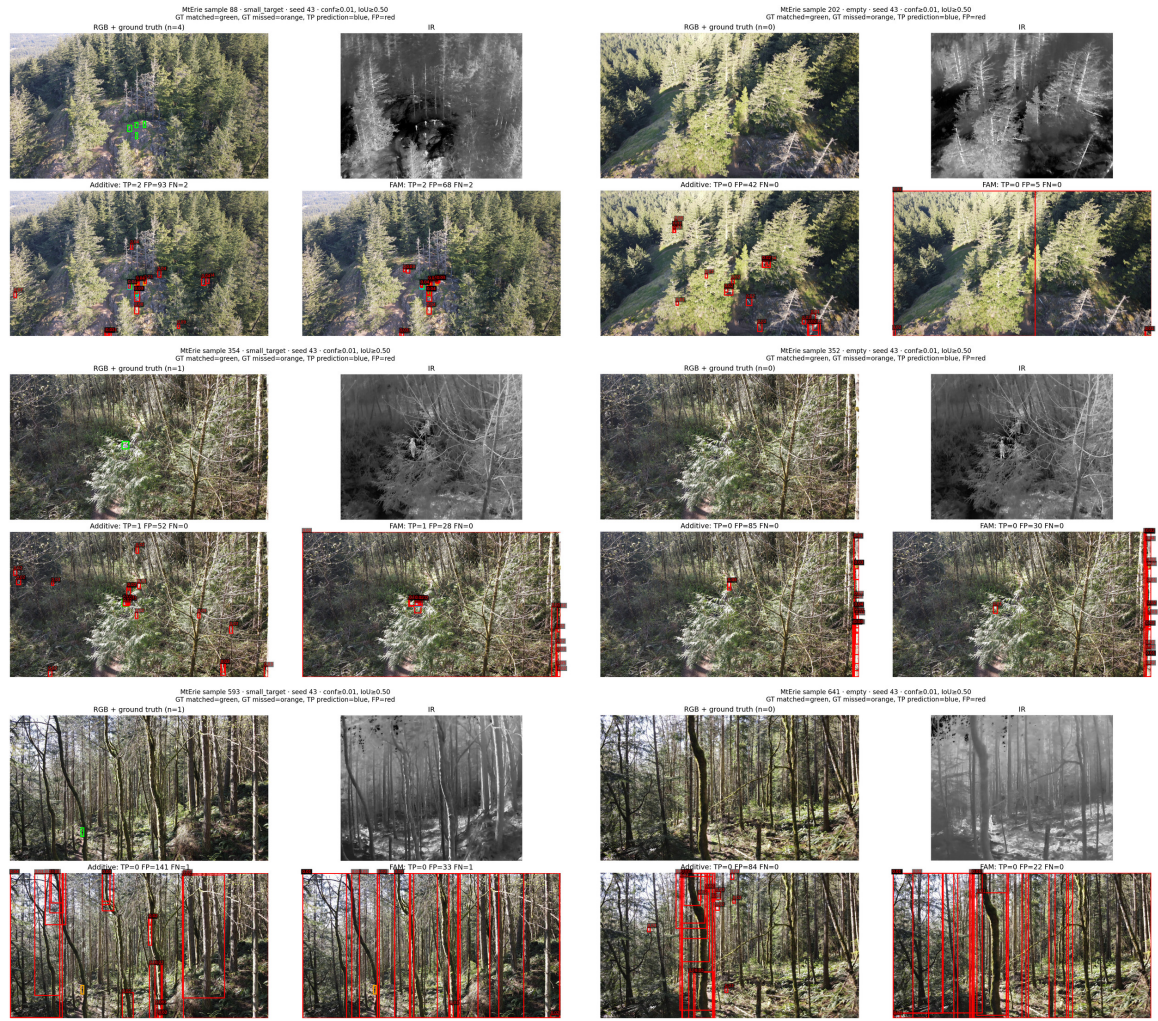


Figure A.1: The six predefined qualitative frames at confidence threshold 0.01. Titles count all detections, while at most twenty high-score predictions are drawn for readability. The density of low-confidence false positives illustrates why 0.01 is a metric-collection threshold rather than an operational setting.

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